AZELCtrl Manual

AZELCtrl — User Manual

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Table of Contents

1. Introduction
   - What is AZELCtrl?
   - The Problem It Solves
   - Target Audience
2. Key Features & Benefits
3. How It Works
4. Technical Details
5. Over-the-Top (ELF) Tracking Mode
6. Menu System Reference
7. Construction Guide
8. Operating the Device
   - Power-Up Sequence
   - Manual Control via Buttons
   - Remote Control via Web Interface
   - Connecting Tracking Software
   - Configuring via OLED Menus
9. Troubleshooting
10. Appendices

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1. Introduction

1.1 What is AZELCtrl?

AZELCtrl is a WiFi-enabled, dual-axis rotator controller designed for amateur satellite operators using Yaesu elevation-azimuth rotator systems. Built around an ESP8266 microcontroller, it provides precise motor control, real-time position feedback, and network connectivity for integration with popular satellite tracking software such as Gpredict and Rotor.

1.2 The Problem It Solves

Yaesu rotator controllers (e.g., G-5600B) provide excellent mechanical performance but lack modern connectivity options. Traditional control requires direct serial connection to a computer, limiting flexibility and remote operation. AZELCtrl bridges this gap by:

- Providing both USB serial and WiFi-based remote control via web browser and TCP/IP protocols
- Enabling seamless integration with satellite tracking software running locally via USB or remotely on the network
- Adding intelligent features such as over-the-top tracking mode for problematic satellite passes
- Offering real-time position monitoring via an integrated OLED display

1.3 Target Audience

This product is designed for amateur radio operators who:

- Own a Yaesu azimuth-elevation rotator system (G-550, G-550X, G-5600, G-5600B, or compatible)
- Track low Earth orbit (LEO) satellites and require automated antenna pointing
- Want flexible control options (USB serial for simple setups, WiFi for remote operation) without replacing existing rotator hardware
- Are comfortable with basic electronics assembly and wiring

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2. Key Features & Benefits

2.1 Core Features

Feature	Description
Dual-Axis Motor Control	Independent control of azimuth and elevation motors with conditional stop logic and direction-reverse safety pause
WiFi Connectivity	ESP8266-based wireless operation — control from any device on the same network
Web Interface	Browser-based control panel with directional buttons and real-time position display
TCP/IP Protocol Support	Compatible with satellite tracking software via TCP port 4533
Multiple Protocol Variants	Supports GS-232A, GS-232B, Easycom, and rotctl text protocol
Real-Time Position Display	128×64 OLED screen showing azimuth, elevation, movement direction, and system status
Potentiometer Calibration	Built-in calibration system with voltage monitoring and safety limits
Over-the-Top (ELF) Mode	Eliminates rotator wrap-around problems during problematic satellite passes
Rotator Model Support	Configurable for N360, S360, N450, and S450 rotator variants

2.2 Benefits Over Traditional Controllers

- **Flexible connectivity** — control via USB serial (simple, direct) or TCP/IP network (remote, flexible)
- **Remote monitoring** — check antenna position from anywhere via web browser when using WiFi mode

- **Multiple control methods** — physical buttons, web interface, USB serial, or TCP/IP network
- **Intelligent tracking** — ELF mode automatically handles passes that would otherwise cause the rotator to perform a damaging full rotation
- **Safety mechanisms** — voltage monitoring protects sensitive electronics; direction-reverse pause prevents motor damage
- **Compact and self-contained** — all components fit in a small enclosure

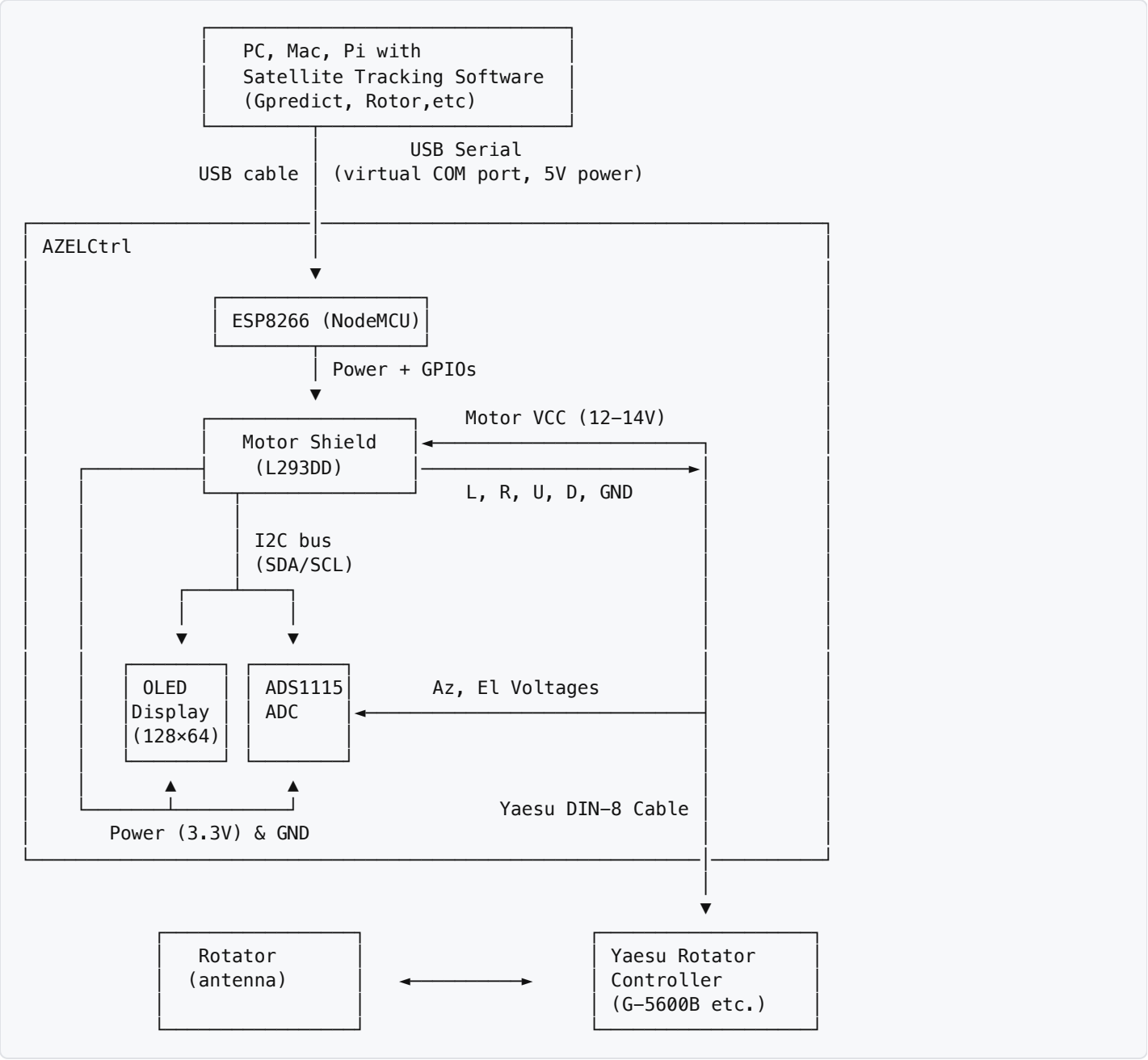
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3. How It Works (High-Level Overview)

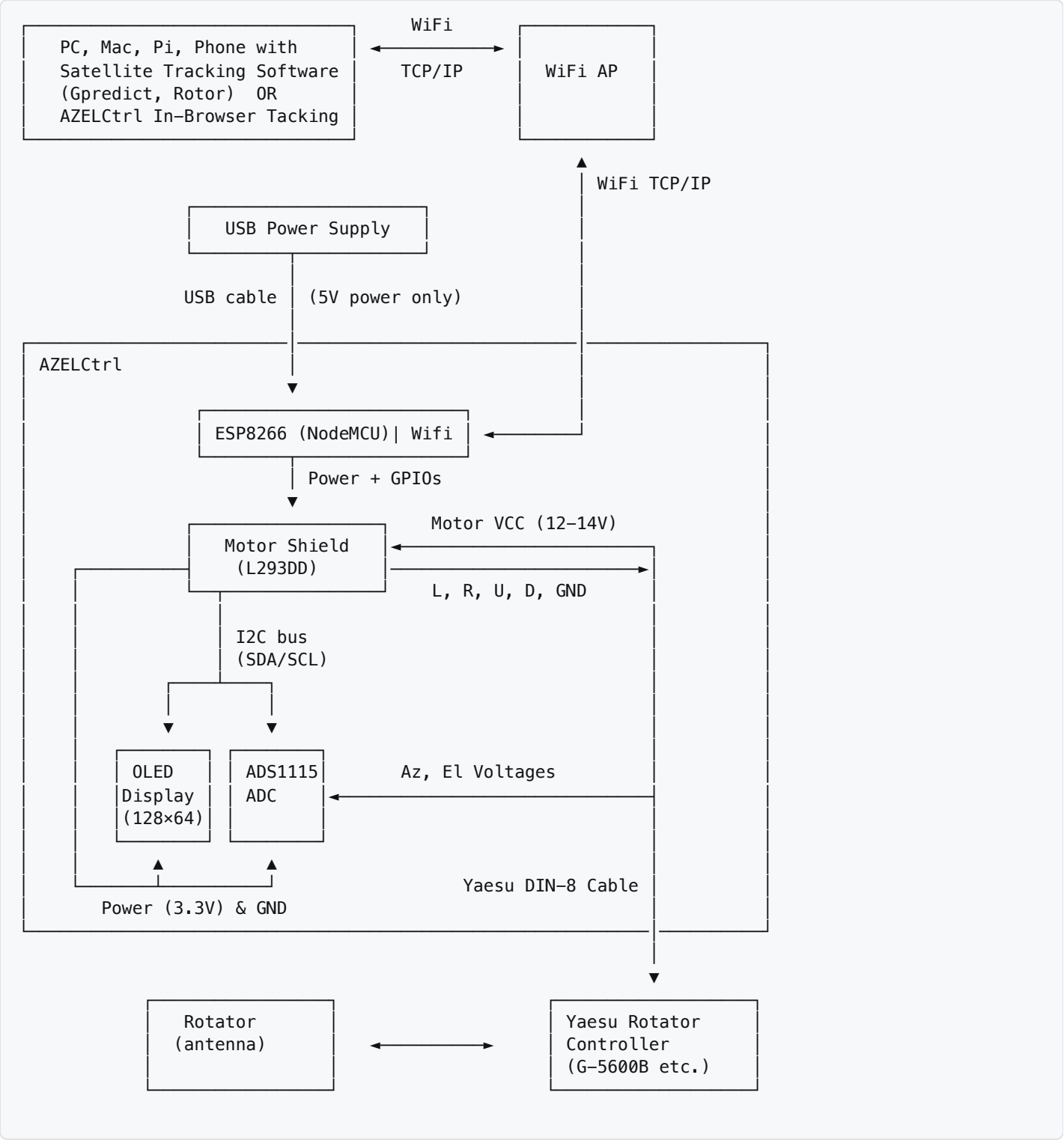
3.1 System Architecture

AZELCtrl sits between your satellite tracking software and your Yaesu rotator controller. All connections between AZELCtrl and the Yaesu controller go through a single 8-wire cable terminating on a DIN-8 connector:

Option 1: USB Serial Connection (Simpler, Direct)



Option 2: WiFi/TCP Connection (Remote, Flexible)



The single Yaesu DIN-8 cable carries all signals:

DIN Pin	Signal	Connection in AZELCtrl
Pin 1	Elevation feedback voltage	ADS1115 A1
Pin 2	Azimuth CW motor control	Motor Shield A+
Pin 3	Elevation Down motor control	Motor Shield B+

DIN Pin	Signal	Connection in AZELCtrl
Pin 4	Azimuth CCW motor control	Motor Shield A-
Pin 5	Elevation Up motor control	Motor Shield B-
Pin 6	Azimuth feedback voltage	ADS1115 A0
Pin 7	+12V power (from Yaesu)	Motor Shield VM
Pin 8	Ground	Motor Shield GND

There are no separate cables for motor control or position sensing — everything goes through this one DIN-8 connector.

Which option should you use?

- **USB Serial** is simpler: one cable provides both power and data. No WiFi configuration needed. Connect AZELCtrl to your computer via USB, select the virtual COM port in your tracking software, and you're ready. This is ideal if your computer is close to the rotator setup.
- **WiFi/TCP** is more flexible: your tracking software can run on any device on the same network — a laptop across the room, a tablet in your shack, or even a phone. AZELCtrl can be powered by a standalone USB power supply (no computer needed). This is ideal for remote operation or when your computer isn't near the rotator.

3.2 Data Flow

AZELCtrl supports two communication interfaces:

USB Serial Interface:

1. **Command Path**: Satellite tracking software sends position commands (azimuth/elevation) over USB serial to AZELCtrl's virtual COM port. The ESP8266 interprets the command according to the configured protocol (GS-232A, GS-232B, or Easycom), calculates the required motor movements, and sends control signals to the motor shield.
2. **Position Feedback**: AZELCtrl reports current position back over USB serial when polled.

TCP/IP Network Interface:

1. **Command Path**: Satellite tracking software sends position commands over TCP to AZELCtrl on port 4533. The protocol spoken is determined by the Hamlib Server setting (Yes = rotctl text protocol, No = configured GS-232/Easycom protocol).
2. **Position Feedback**: AZELCtrl reports current position back over TCP when polled.

Motor Control Path (same for both interfaces): The motor shield (L293DD) receives direction and enable signals from the ESP8266 and switches the Yaesu controller's relay inputs to ground, activating the appropriate rotator motors.

Position Feedback Path (same for both interfaces): The Yaesu controller outputs analog voltages proportional to the current azimuth and elevation positions. These voltages are read by the ADS1115 ADC module, converted to digital values by the ESP8266, and displayed on the OLED screen. Position data is also reported back to the tracking software when polled, via whichever interface is in use.

3.3 Power Flow

- AZELCtrl is powered via USB (5V to VIN) or an external 5V supply.
- Motor drive voltage (12V) is supplied by the Yaesu controller through the DIN-8 connector (Pin 7 to VM).
- **Critical**: VIN (5V logic) and VM (12V motor) must remain electrically isolated. The motor shield's VIN-VM jumper must be removed.
- **Rotor power detection** (v1.5.11, optional): AZELCtrl can detect whether the Yaesu controller is connected and powered, by monitoring the +12 V rail on Pin 7 through a voltage divider on the ADS1115's free A2 input. Fit a **1.3 MΩ** resistor from the motor shield's **VM** pin to the ADS1115's **A2** pin, and a **200 kΩ** resistor from **A2** to **GND**. This scales the rail to ~1.6 V at 12 V nominal — and only ~2.7 V even during the Yaesu's poorly regulated 20 V spikes, safely inside the ADS1115's absolute-max input rating. With the controller unplugged or unpowered, A2 sits near 0 V. The firmware debounces the reading (2.5 s) with hysteresis, so motor-load sags and plug transients cannot flap the state. Enable or disable it in Settings → Rotor Power Detection (on by default). **Until the resistors are fitted, keep this setting OFF** — an unwired A2 input floats and gives meaningless readings.

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4. Technical Details

4.1 Hardware Components

Component	Specification	Purpose
ESP8266 NodeMCU v2	80 MHz, 4 MB Flash, built-in WiFi	Main microcontroller — runs firmware, handles protocols, WiFi, motor control
ESP-12E Motor Shield	L293DD dual full-bridge driver	Provides motor control outputs and I/O expansion headers
SSD1306 OLED Display	128×64 pixels, I2C interface	Real-time position display and menu system
ADS1115 ADC Module	16-bit, 4-channel, I2C interface	Converts Yaesu potentiometer voltages to digital readings
Push Buttons (×2)	Momentary, normally open	Physical menu navigation and manual control
Enclosure	Small plastic project box	Houses all components securely

4.2 Pin Assignments

ESP8266 → Motor Shield (Hard-Wired)

NodeMCU Pin	GPIO	Function	Connected To
D1	GPIO5	Motor A Enable	Azimuth motor enable
D3	GPIO0	Motor A Direction	Azimuth direction (CW/CCW)
D2	GPIO4	Motor B Enable	Elevation motor enable
D4	GPIO2	Motor B Direction	Elevation direction (Up/Down)

ESP8266 → External Components

NodeMCU Pin	GPIO	Function	Connected To
D6	GPIO12	I2C SDA	OLED SDA + ADS1115 SDA
D7	GPIO13	I2C SCL	OLED SCL + ADS1115 SCL
D8	GPIO15	Button 1 (active HIGH)	Btn1 → 3.3V (GPIO15 has internal pull-down)
D5	GPIO14	Button 2 (active LOW)	Btn2 → GND (GPIO14 has internal pull-up)

Motor Shield → Yaesu DIN-8

Shield Terminal	Yaesu DIN Pin	Function
A+	Pin 2	Azimuth Clockwise
A-	Pin 4	Azimuth Counter-Clockwise
B+	Pin 3	Elevation Down
B-	Pin 5	Elevation Up
VM	Pin 7	12V from Yaesu controller
GND	Pin 8	Common ground

ADS1115 → Yaesu DIN-8 (Position Feedback)

ADS1115 Channel	Yaesu DIN Pin	Function
A0	Pin 6	Azimuth position voltage
A1	Pin 1	Elevation position voltage

4.3 Motor Control Theory

The L293DD motor driver is configured to control the Yaesu controller's relay inputs. Each Yaesu relay is activated by grounding its control pin. The motor shield provides three states per output:

State	Shield Pin +	Shield Pin -	Yaesu Relay
Open circuit	Floating	Floating	De-energized (motor stopped)
Ground	VCC	GND	Energized (motor active)
High voltage	GND	VCC	De-energized (12V ≈ open circuit for Yaesu)

The ESP8266 controls direction by selecting which motor shield output is grounded, and enables/disables movement by toggling the enable pin. The shield's design prevents both inputs from being grounded simultaneously, even in the event of a programming error.

4.4 I2C Bus Architecture

Both the OLED display and ADS1115 share the I2C bus:

- **SDA**: NodeMCU D6 (GPIO12)
- **SCL**: NodeMCU D7 (GPIO13)
- **OLED address**: 0x3C

- **ADS1115 address**: 0x48 (default, determined by address jumpers)

The ADS1115 operates at 16-bit resolution with a programmable gain amplifier. For our application, the ±4.096V range is used to safely accommodate the Yaesu controller's output voltages (which must be trimmed to ≤3.3V).

4.5 Communication Interfaces and Protocol Stack

AZELCtrl supports two communication interfaces:

USB Serial Interface

When AZELCtrl is connected to a computer via USB, it appears as a **virtual COM port** (e.g., COM3 on Windows, /dev/cu.usbserial-* on macOS, /dev/ttyUSB* on Linux). The ESP8266's built-in USB-to-serial converter (CP2102) handles the conversion transparently.

- **Protocol**: The configured serial protocol is used (GS-232A, GS-232B, or Easycom — selected in the PC Interface submenu under SETUP).
- **Baud rate**: 115200 baud (default).
- **Power**: The USB connection provides 5V power to AZELCtrl, so no separate power supply is needed.

This interface is ideal when:

- Your computer is physically close to the rotator setup.
- Your tracking software only supports serial connections (some older or simpler programs).
- You want the simplest possible setup with no WiFi configuration.
- Your tracking software has issues with TCP/IP but works reliably over serial.

TCP/IP Network Interface

AZELCtrl runs a TCP server on **port 4533** that accepts connections from satellite tracking software. The protocol spoken over TCP is determined by the **Hamlib Server** setting in the rotator menu:

Hamlib Server	TCP Protocol
Yes	rotctl text protocol (`p` , `P` , `S` commands)
No	Configured serial protocol (GS-232A, GS-232B, or Easycom)

The server always listens on port 4533 regardless of the Hamlib Server setting — the setting only determines which protocol dialect is used for communication.

This interface is ideal when:

- Your tracking software runs on a different device (laptop, tablet) on the same network.
- You want remote operation without a physical cable to the rotator.
- AZELCtrl is powered by a standalone USB power supply (no computer connection needed).

Important: Both interfaces can be used simultaneously. AZELCtrl processes commands from whichever interface they arrive on. However, having two programs sending commands at the same time (one over USB, one over TCP) will cause conflicts — only use one control source at a time.

GS-232 Command Reference

Command	Description	Response
`C`	Query current position	`Caaa.ee\r` (AZ=aaa, EL=ee)
`Maaa`	Move azimuth to aaa degrees	`OK\r`
`Waaa eee`	Move to azimuth aaa, elevation eee	`OK\r`
`L`	Move azimuth left (CCW)	`OK\r`
`R`	Move azimuth right (CW)	`OK\r`
`U`	Move elevation up	`OK\r`
`D`	Move elevation down	`OK\r`
`S`	Stop all movement	`OK\r`

rotctl Command Reference

Command	Description	Response
`p`	Query position	`az\nel\nRPRT 0\n`
`P az el`	Move to position	`RPRT 0\n`
`S`	Stop	`RPRT 0\n`

4.6 Coordinate Transformations

Rotator Model Variants

AZELCtrl supports four rotator model configurations:

Model	Range	Wrap Boundary	Typical Use
N360	0°–360° (North-centered)	0°/360° (North)	Southern Hemisphere
S360	0°–360° (South-centered)	180° (South)	Northern Hemisphere
N450	0°–450° (North-centered)	0°/360° (North)	Southern Hemisphere, extended range
S450	0°–450° (South-centered)	180° (South)	Northern Hemisphere, extended range

The selected model affects how the firmware interprets raw ADC readings and calculates shortest-path movement directions.

Auto-Stop Logic

AZELCtrl implements mechanical offset tracking for extended-range models:

- **N360**: Auto-stops at 180° (South)
- **S360**: Auto-stops at 0° (North)
- **N450**: Auto-stops at 135° and 225° (overlap zone)
- **S450**: Auto-stops at 315° and 45° (overlap zone)

A configurable reverse delay (0–3 seconds, default 1 second) is applied when changing direction to allow motor momentum to settle.

Preventing Tracking Oscillations

When AZELCtrl receives position targets from tracking software, it must decide when the rotator has reached its destination and stop the motors. This is not as simple as it sounds due to mechanical inertia.

The problem: When motor power is cut, the rotator does not stop instantly. Mechanical momentum carries it past the target point before friction brings it to rest. If AZELCtrl waits until the measured position matches the target exactly before cutting power, the rotator will consistently overshoot by several degrees. The tracking software then sends a correction command to move back toward the target, the rotator overshoots again in the opposite direction, and the system enters a continuous ±3–5° oscillation.

The solution: AZELCtrl uses a **3.0° auto-stop tolerance** for protocol-controlled tracking. When the current position is within 3.0° of the target, the motors are powered off early enough that mechanical inertia carries the rotator to within the target zone rather than past it. This effectively "predicts" the overshoot and compensates for it.

The position feedback loop runs at 100ms intervals (ADC reads every 100ms, auto-stop checks every 100ms), giving AZELCtrl timely awareness of the rotator's position while avoiding excessive CPU load. The 3.0° tolerance was chosen empirically: it is large enough to prevent oscillation during normal tracking speeds, yet small enough to maintain acceptable pointing accuracy for satellite contacts.

4.7 EEPROM Layout

Configuration data is persisted in EEPROM across power cycles:

Bytes	Size	Content	Magic
0–66	67	WiFi SSID (data at 1–32) and password (data at 34–65)	0xAA at byte 66
67–83	17	Calibration min/max voltages (4× float: Az min, Az max, El min, El max)	0xCA at byte 83
84–85	2	WiFi mode (STA/AP/Disabled)	0x4D at byte 85
86–87	2	Network settings (Hamlib Server toggle)	0x4E at byte 87
88–89	2	Rotator model (N360/S360/N450/S450)	0x5A at byte 89
90	1	ELF mode enabled/disabled	—
91–99	9	AP password (8 chars)	0x41 ('A') at byte 99
100–104	5	Protocol settings — baud index, show commands, reverse delay, protocol variant	0xB7 at byte 102

4.8 Safety Mechanisms

- **Direction-reverse pause**: 600ms pause when reversing motor direction to allow momentum to settle. Per-axis independent — reversing azimuth while elevation is idle only pauses azimuth.
- **Voltage monitoring**: During calibration, voltages are checked every ~100ms. Warning at 1.65V, hard stop at 3.3V (ADS1115 damage threshold).
- **Conditional motor stops**: Motors are only stopped when changing direction or stopping entirely — not when continuing in the same direction. This prevents unnecessary engagement delays that could cause tracking software timeouts.
- **Engagement delay**: 200ms delay after motor re-engagement to mask mechanical backlash without triggering client timeouts.
- **Safe boot state**: On power-up, all motors are disabled. No movement occurs until explicitly commanded.

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5. Over-the-Top (ELF) Tracking Mode

5.1 The Wrap-Around Problem

Amateur satellite operators routinely track low Earth orbit (LEO) satellites using azimuth-elevation rotator systems. In most passes, the rotator simply follows the satellite across the sky without incident. However, a particular class of pass presents a serious mechanical challenge: those in which the satellite crosses the rotator's **wrap-around boundary** — the azimuth heading at which the rotator must reverse direction or rotate nearly a full 360 degrees to continue tracking.

Consider a Southern Hemisphere operator using a north-centered rotator. A satellite rises in the southwest at Az=225°, climbs to a peak near Az=180° (due south), and sets in the southeast at Az=135°. The rotator tracks perfectly until the satellite approaches 180°. To continue tracking, the controller would need to move from, say, Az=181° to Az=179° — but the rotator's mechanical range ends at ±180°. To reach 179°, the drive motor must reverse direction and travel nearly 360° clockwise (past north, east, south, and back). On a typical rotator with a slew speed of around 4 degrees per second, this journey takes well over a minute.

During that minute, the antenna is pointed nowhere near the satellite. For a voice or data contact, this is catastrophic. The link drops, the pass is lost, and the opportunity is wasted.

The equivalent problem for Northern Hemisphere operators occurs when a satellite passes to their north — rising at Az=330° and setting at Az=90°, crossing through Az=0°/360°. The mechanics are identical, and the solution is the same.

5.2 The Geometric Principle

Most amateur-grade elevation rotators designed for satellite work support elevation movement not just from 0° (horizon) to 90° (zenith), but all the way to 180°. An elevation of, for example, 135° means the antenna is tilted past vertical and is now pointing toward a direction 45° above the horizon on the **opposite side** of the rotator.

This mechanical capability is the key to the over-the-top technique. We refer to this as ELF (ELevation Flip) mode.

The insight is straightforward: any point in the sky can be reached from two different azimuth headings. If a satellite is at Az=180°, El=35°, you can point your antenna at it directly — or you can point your antenna at Az=0°, El=145°. In the latter case, the antenna is rotated to face north, but elevated past vertical so that it points back over the top toward the south at an angle of 35° above the southern horizon.

The transformation is:

```
Real Azimuth   = (Satellite Azimuth + 180°) mod 360°
Real Elevation = 180° − Satellite Elevation
```

Applied continuously during the pass, this means the azimuth rotator is smoothly tracking through the opposite side of sky — through its comfortable travel range near Az=0° (or Az=180° for the Northern Hemisphere case) — while the elevation rotator holds the antenna pointed at the correct elevation from the other side.

5.3 How AZELCtrl Implements ELF Mode

AZELCtrl implements ELF mode as a transparent coordinate transformation layer. When enabled:

1. **Position reports**: When tracking software polls for position (`C` or `p` command), AZELCtrl reads the physical position from the ADC, applies the inverse ELF transformation, and reports the **effective** (satellite-space) coordinates. The tracking software sees the antenna pointing directly at the satellite, even though the rotator is physically in the opposite direction.
2. **Movement commands**: When tracking software sends a movement command (`W`, `P`, `M`), AZELCtrl receives the satellite-space target, transforms it to physical coordinates, and steers the rotator to the translated position.
3. **Direction calculation**: All shortest-path direction calculations are performed in consistent coordinate space to ensure correct movement direction regardless of ELF mode state.
4. **Elevation prompt**: When ELF mode is toggled, AZELCtrl displays an "Elevate To?" prompt allowing the operator to immediately set the effective elevation to Horizon (0°), 45°, 90°, or Skip (no change). This is useful for quickly repositioning the antenna before a pass.

5.4 When to Use ELF Mode

The over-the-top transformation should be applied when:

- The satellite pass crosses the rotator's wrap-around boundary
- The pass cannot be tracked in direct mode without the rotator performing a full rotation

The over-the-top transformation should **not** be applied when:

- The pass does not cross the wrap-around boundary (unnecessary mechanical load on the elevation rotator)
- Linear polarisation must be maintained without adjustment (over-the-top mode rotates the polarisation plane by 180°)

5.5 Practical Considerations

- **Elevation rotator capability**: Verify that your elevation rotator physically supports movement beyond 90°. The Yaesu G-5600B, for example, supports 0–180° of elevation travel.
- **Pre-positioning before AOS**: For over-the-top passes, the antenna must be pre-positioned to the starting position before the satellite rises. This typically means the elevation will be near 180° while waiting for AOS. The firmware includes a built-in web-based tracking feature that automatically determines when ELF mode is required. When the user clicks the Start button to begin tracking the selected satellite, the system prompts the user if ELF mode needs to be enabled.
- **Polarisation effects**: Flipping the antenna over the top rotates the polarisation plane by 180°. For circular polarisation this is irrelevant. For linear polarisation, you may need to adjust the polarisation rotator or accept a polarisation mismatch.
- **Decision timing**: The decision to use over-the-top mode must be made **before** the pass begins, based on a pre-computed prediction of the complete azimuth track. Switching modes mid-pass would require a large, sudden azimuth slew that would lose the satellite.

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6. Menu System Reference

6.1 Button Behavior Summary

Button	Pin	Electrical Behavior
Btn1	D8 (GPIO15)	Active HIGH — pulls to 3.3V when pressed (internal pull-down)
Btn2	D5 (GPIO14)	Active LOW — pulls to GND when pressed (internal pull-up)

Note on GPIO15: This pin must be LOW at reset for the ESP8266 to boot normally. The motor shield comes with a fixed pull-down resistor. If you power up the unit while Btn1 is pressed, it will not boot properly.

Button actions in rotator control mode:

Action	Effect
Btn1 short-press	Cycle through: SETUP → LEFT → RIGHT → UP → DOWN → ELF
Btn2 press (hold)	Start movement in selected direction (LEFT/RIGHT/UP/DOWN). Release to stop.
Btn2 press (when SETUP selected)	Enter SETUP menu
Btn2 press (when ELF selected)	Toggle ELF mode On/Off, shows "Elevate To?" prompt
Btn1+Btn2 long-press (when SETUP selected)	Reboot device

Button actions in menus:

Action	Effect
Btn1 short-press	Move cursor down one item (wraps to top)
Btn1 long-press	Select current item
Btn2	Go back to parent menu

Understanding the menu footer:

Every menu screen displays a footer line that tells you which button action to use. The format is:

Symbol	Meaning
`1.`	Short-press Btn1 (tap and release quickly)
`1|`	Long-press Btn1 (press and hold for ~1 second)
`1!`	Hold Btn1 (press and continue holding until action completes)
`2.`	Short-press Btn2
`2|`	Long-press Btn2
`2!`	Hold Btn2

The separator between actions is a space and a period (`.`) or a pipe (`|`). For example:

- `1.Nav 1|Sel 2.Back` means: short-press Btn1 to navigate, long-press Btn1 to select, short-press Btn2 to go back
- `1.OK 2.Back` means: short-press Btn1 to confirm, short-press Btn2 to go back

- `1.Done 2.Cancel` means: short-press Btn1 to finish, short-press Btn2 to cancel

The footer adapts to the current screen's available actions.

6.2 Startup Screens

Splash Screen

On power-up, AZELCtrl displays a splash screen for 3 seconds:

```
AZELCtrl

Ver: 1.5.1
Date: Aug 1 2026
By: ZS6TW
```

Shows product name, firmware version, build date, and author.

Missing Settings

If mandatory settings are incomplete (rotator variant not configured, calibration not done, or WiFi mode not set), a notice appears for 3 seconds:

```
Missing
Settings!
```

After the notice, the device goes directly to the SETUP menu. The footer omits the Back button — you must complete the mandatory settings before leaving.

Boot Menu

If all mandatory settings are configured, the Boot menu appears:

```
Boot
————
Rotator Ctrl
Setup

. . . . . .
1.Nav 1|Sel
```

- **Rotator Ctrl**: Starts the rotator controller. Has a 3-second countdown — auto-selects if no button is pressed.
- **Setup**: Opens the SETUP menu.

Button behavior: Btn1 short-press cycles options. Btn1 long-press selects. There is no Back button (this is the root menu).

WiFi Connecting

When "Rotator Ctrl" is selected and WiFi mode is set to Station, the WiFi connecting screen appears:

```
WiFi Connect

SSID: YourWifiSSID
Waiting: 3s
```

Shows the SSID being connected to and a running elapsed time counter. The device attempts to connect for up to 15 seconds.

WiFi Connected

On successful connection:

```
Connected!

IP: 192.168.1.42
Signal: -65 dBm
SSID: YourWifiSSID
                    3
```

Shows the assigned IP address, signal strength, and SSID. A countdown timer in the bottom-right corner counts down from 5 seconds. Press any button to skip the countdown and proceed immediately to the rotator control screen.

WiFi Failed

If the connection times out:

```
WiFi
Failed

SSID: YourWifiSSID

Status: WL_CONNECT_FAILED
```

Shows the SSID that failed and the WiFi status code. Displays for 2 seconds, then proceeds to the rotator control screen without WiFi. The device can still be controlled via USB serial or through the OLED menu.

6.3 Main Az/El Control Screen

This is the primary screen shown when the rotator controller is running. It has no title bar — the full display is used for position values.

Normal idle state (no movement, no protocol commands):

```
Az:  125.3d →
El:   45.2d ↑

Sel: LEFT    ELF

. . . . . . . . . . . . . . . . . .
1.Cycle 2!Go
```

Line	Content
1	Azimuth position (large font) with optional movement arrow (→ for CW, ← for CCW) on the right
2	Elevation position (large font) with optional movement arrow (↑ for UP, ↓ for DOWN) on the right
3	When idle: `Sel:` followed by the currently selected direction (SETUP/LEFT/RIGHT/UP/DOWN). When ELF mode is enabled, `ELF` appears right-aligned on this line.

Line	Content
4	Dotted separator line above the footer
5	Button instructions footer

When moving (but no protocol commands showing):

```
Az:  125.3d
El:   45.2d

Moving: RIGHT

. . . . . . . . . . . . . . . . . .
1.Cycle 2!Go
```

Line 3 shows `Moving: direction` instead of `Sel:` while motors are active.

When receiving protocol commands (if Show Commands is enabled):

```
Az:  125.3d
El:   45.2d

. . . . . . . . . . . . . . . . . .
SR: C
ST: AZ=125 EL=045
```

Lines 3-4 show the last received command and response. The two-letter tag encodes source + direction: first letter **S** = USB serial, **N** = network (TCP); second letter **R** = received, **T** = transmitted (so `SR:` = received over serial, `NT:` = transmitted over network). Replies are sent back over the same interface as the command, so both lines share the same source letter. Newlines inside multi-line responses are shown as `^`, and lines longer than the display width are truncated. This overlay is suppressed for 5 seconds after any button press so the menu stays visible during interaction.

SAFETY PAUSE screen (shown when reversing direction):

```
SAFETY PAUSE
────────────
AZ axis
125.3d El: 45.2d

. . . . . . . . . . . . . . . . . .
1.Cycle 2!Go
```

Title bar shows "SAFETY PAUSE" with which axis is pausing. Press Btn2 to skip the pause and continue immediately.

TCP Connection Indicator: When a tracking software is connected via TCP, a small white dot appears in the top-right corner of the screen.

Footer varies by selection:

- SETUP selected: `1.Cycle 2!Go 1+2!|RST` (1+2 long-press reboots)
- ELF selected (on): `1.Cycle 2!Off`

- ELF selected (off): `1.Cycle 2!On`
- Direction selected: `1.Cycle 2!Go`

Movement arrows: Arrows on the Az/El lines indicate which axis is currently moving:

- Azimuth: → (right arrow) for CW, ← (left arrow) for CCW
- Elevation: ↑ (up arrow) for UP, ↓ (down arrow) for DOWN
- In ELF mode, elevation arrows reflect **effective** satellite direction (swapped from physical motor direction)
- Arrows disappear when the axis is idle

ELF Indicator: When ELF mode is enabled, "ELF" is displayed right-aligned on the `Sel:` line.

6.4 SETUP Menu

Accessed from the Boot screen by selecting "Setup", or from the main Az/El control screen by cycling to "SETUP" and long-pressing Btn2.

```
SETUP
-----
Rot Variant (N360)
Rot Calibrate (Y)
Network
PC Interface
Wipe all settings

. . . . . .
1.Nav  1|Sel  2.Back
```

Option	Function
Rot Variant	Opens rotator model selection screen to choose N360, S360, N450, or S450
Rot Calibrate	Opens the Voltage Calibration menu for azimuth and elevation potentiometers
Network	Opens Network submenu (see below)
PC Interface	Opens PC Interface submenu (see below)
Wipe all settings	Resets all configuration to factory defaults (with confirmation)

Labels update dynamically:

- **Rot Variant**: Shows current model in parentheses when configured (e.g., "Rot Variant (N360)"), or just "Rot Variant" if not yet set.
- **Rot Calibrate**: Shows "(Y)" if both axes are calibrated, "(N)" if either axis is uncalibrated.
- **Back button**: Hidden until all mandatory settings are configured (rotator model selected, both axes calibrated, WiFi mode set).

Button behavior: Btn1 short-press cycles through options. Btn1 long-press selects the current option. Btn2 goes back to the previous screen.

Rotator Variant Selection Screen

Selecting "Rot Variant" opens a dedicated model selection screen:

```
Rotator Variant?
----------------
N360 (-180 to 180)
S360 (0 to 360)
N450 (-225 to 225)
S450 (-45 to 405)

. . . . . . . . . . . . . . . . .
1.Nav  1|Sel  2.Back
```

Understanding the model names:

- The first letter indicates the direction the rotator points when the azimuth is at the center of its range: **N** = North, **S** = South.
- The number indicates the total azimuth span the rotator is designed to handle: **360** = 360° of travel, **450** = 450° of travel (extended range with an additional 90° beyond standard).

Model	Range	Wrap Boundary	Typical Use
N360	-180° to +180°	0°/360°	Southern Hemisphere
S360	0° to 360°	180° (South)	Northern Hemisphere
N450	-225° to +225°	0°/360°	Southern Hemisphere, extended range
S450	-45° to +405°	180° (South)	Northern Hemisphere, extended range

Controls:

- Btn1 short-press: Cycle through models (resets any auto-select timer)
- Btn1 long-press: Select and save the highlighted model
- Btn2: Go back without changing

Important: Changing the rotator model clears the existing calibration values. You must recalibrate after selecting a new model.

Voltage Calibration Menu

Selecting "Rot Calibrate" opens the calibration menu:

```
Voltage Calibration
-------------------
Azimuth
Elevation
Skip (Az=El=3.2V)

. . . . . . . . . . . . . . . . .
1.Nav  1|Sel  2.Back
```

Option	Function
Azimuth	Start azimuth potentiometer calibration (sweep and return sequence)

Option	Function
Elevation	Start elevation potentiometer calibration (sweep and return sequence)
Skip (Az=El=3.2V)	Set both axes to fixed voltage ranges (min=0V, max=3.2V) without automatic calibration

Controls:

- Btn1 short-press: Cycle through options
- Btn1 long-press: Select current option
- Btn2: Go back to SETUP menu

See section 6.6 for details on the calibration screens.

Network Submenu

Selecting "Network" opens the network settings menu:

```
Network
-------
WiFi Mode(Station)
Hamlib Server (Y)

. . . . . .
1.Nav  1|Sel  2.Back
```

Option	Function
WiFi Mode(label)	Opens WiFi Mode submenu to select Station, Access Point, or Disable. Label reflects current mode: `WiFi Mode(None)` (not set), `WiFi Mode(Station)`, `WiFi Mode(AP)`, or `WiFi Mode(Disabled)`
Hamlib Server	Toggle Yes/No — determines TCP protocol: Yes = rotctl text protocol, No = configured serial protocol (GS-232A, GS-232B, or Easycom). Footer shows current toggle action (e.g., `1.Nav 1

Controls:

- Btn1 short-press: Cycle through options
- Btn1 long-press: Select current option (or toggle Hamlib Server when on that line)
- Btn2: Go back to SETUP menu

WiFi Mode Submenu

Selecting the WiFi Mode line opens the WiFi Mode submenu:

```
WiFi Mode
---------
Station
Access Point
Disable

. . . . . .
1.Nav  1|Sel  2.Back
```

Option	Function
Station	Sets WiFi mode to Station and opens the Station Settings screen

Option	Function
Access Point	Sets WiFi mode to Access Point and opens the AP Mode menu
Disable	Disconnects from any network, turns off WiFi, clears saved credentials, and returns to SETUP

Controls:

- Btn1 short-press: Cycle through options
- Btn1 long-press: Select current option
- Btn2: Go back to Network menu

Station Settings Screen

Selecting "Station" opens the Station Settings screen:

```
Wifi Station Settings
---------------------
Set via HTTP
Connect

SSID: YourNetworkName
Passwd: a*..*z

. . . . . . . . . . . . . . . . . .
1.Nav  1|Sel  2.Back
```

Option	Function
Set via HTTP	Starts a captive portal (SSID: AZELCTRL_XXXX, IP: 192.168.4.1). Connect your phone/laptop to the device network and open a browser to configure your home WiFi credentials
Connect	Connects to the saved SSID. Label changes to "Disconnect" when connected

Information displayed:

- **SSID**: Shows the saved SSID, or "None" if no credentials are saved
- **IP**: Shows the assigned IP address when connected
- **Passwd**: Shows a masked password (first and last character visible) when credentials are saved but not connected

Controls:

- Btn1 short-press: Cycle through options
- Btn1 long-press: Select current option
- Btn2: Go back to SETUP menu

AP Mode Menu

Selecting "Access Point" opens the AP Mode menu:

```
AP Mode
_______
AP Status
Regen AP Password

. . . . . . . . . . . . . . . . . .
1.Nav  1|Sel  2.Back
```

Option	Function
AP Status	Show AP information (SSID, password, IP)
Regen AP Password	Generate a new random 8-character alphanumeric password for AP access

Controls:

- Btn1 short-press: Cycle through options
- Btn1 long-press: Select current option
- Btn2: Go back to WiFi Mode menu

AP Mode: When enabled, the device creates a WiFi access point with SSID "AZELCTRL_XXXX" where XXXX is the last 4 hex digits of the device MAC address. This allows distinguishing multiple devices. By default, the AP password is "yaesu123". You can generate a new random 8-character password by selecting "Regen AP Password". The process:

1. Screen shows "Generating new password..."
2. Press Btn1 to provide timing entropy for randomness
3. New 8-character password is displayed — note it down!
4. Press Btn1 to acknowledge and return to AP Mode menu

Note: WPA2 requires a minimum of 8 characters for the password.

AP Status screen shows:

```
AP Status
_________
AP Mode Active
SSID: AZELCTRL_4D49A1
Pass: yaesu123
IP: 192.168.4.1

. . . . . . . . . . . . . . . . . .
1.OK 2.Back
```

The SSID shown is the actual SSID the device is broadcasting (e.g., AZELCTRL_4D49A1). Connect to this network using the password shown. When connected, the browser automatically opens to the Rotator Control web interface. Press Btn1 to return to AP Mode menu.

PC Interface Submenu

Selecting "PC Interface" opens the PC interface settings menu:

```
PC Interface
------------
Baud Rate: 115200
Show Commands: Yes
Reverse Delay: 1s
Protocol (GS-232B)

. . . . . . . . . . . . . . . . .
1.Nav  1|Sel  2.Back
```

Option	Function
Baud Rate	Opens submenu to select USB serial baud rate: 115200, 57600, 38400, 19200, 9600
Show Commands	Toggle Yes/No — show/hide RX/TX overlay on OLED when protocol commands are received. Footer shows current toggle action (e.g., `1.Nav 1
Reverse Delay	Cycle through 0–3 seconds — delay before reversing motor direction to allow momentum to settle
Protocol	Opens submenu to select GS-232A, GS-232B, or Easycom

Controls:

- Btn1 short-press: Cycle through options
- Btn1 long-press: Select current option (or toggle when on Show Commands line)
- Btn2: Go back to SETUP menu

Baud Rate Selection

Selecting "Baud Rate" opens:

```
Select Baud Rate
----------------
115200
57600
38400
19200
9600

. . . . . . . . . . . . . . . . .
1.Nav  1|Sel  2.Back
```

Selecting a baud rate applies it immediately and returns to the PC Interface menu.

Protocol Selection

Selecting "Protocol" opens:

```
Protocol
--------
GS-232A (Yaesu)
GS-232B (Hamlib)
Easycom

. . . . . . . . . . . . . . . . .
1.Nav  1|Sel  2.Back
```

Selecting a protocol applies it immediately and returns to the PC Interface menu.

Wipe All Settings

Selecting "Wipe all settings" shows a confirmation screen:

```
Wipe all settings
-----------------
This will erase ALL
saved settings.

Continue?

. . . . . . . . . . . . . . . . .
1.Wipe  2.Back
```

Button	Action
Btn1	Erases all EEPROM settings, resets in-memory state, shows "Settings Wiped!" confirmation for 2 seconds, then reboots the device
Btn2	Cancel and return to SETUP menu

What is erased: WiFi credentials, WiFi mode, calibration values, rotator model, Hamlib Server setting, PC Interface settings (baud rate, protocol, etc.). After reboot, the device enters the SETUP menu with all settings at factory defaults.

6.5 ELF Mode Toggle

ELF (over-the-top tracking) mode is toggled from the main Az/El control screen, not from a settings menu.

1. Press Btn1 repeatedly to cycle through options until "ELF" is selected.
2. Press Btn2 to toggle ELF mode On/Off.
3. The "Elevate To?" prompt appears, allowing you to immediately set the effective elevation or skip.

When ELF mode is enabled, "ELF" is displayed on the right side of the OLED screen next to the selected direction indicator.

6.6 Calibration Screens

Accessed from SETUP by selecting "Rot Calibrate". Opens the calibration menu with options for Azimuth, Elevation, or Skip.

Calibration Menu

```
Voltage Calibration
-------------------
Azimuth
Elevation
Skip (Az=El=3.2V)

. . . . . . . . . . . . . . . . . .
1.Nav  1|Sel  2.Back
```

Azimuth/Elevation Calibration Screen

When calibration is running:

```
Cal-AZ: Sweep CCW
Moving CCW...
 AZ:  -180.0 ( 123mV)
 EL:    45.2 (2400mV)
. . . . . . . . . . . . . . . . . .
1.Exit 2.Back
```

Title bar: Shows axis being calibrated (AZ or EL) and current state:

- `Sweep CCW` / `Sweep DOWN` — Moving to find minimum voltage
- `Sweep CW` / `Sweep UP` — Moving to find maximum voltage
- `Return CCW` / `Return DOWN` — Returning while pausing at cardinal headings
- `Adjust Max` — Waiting for operator to position rotator exactly at 180°
- `CALIBRATED` — Calibration complete

Status area: Context-sensitive messages:

- During sweep: `Moving CCW...` or `Moving DOWN...`
- During return: Shows next heading to pause at (e.g., `Next: EAST`) or countdown when paused (e.g., `Stopped at: SOUTH [3]`)
- During adjust: `Adjust Az to exactly 180 deg`

Readings area: Always shows both Az and El positions with voltages in millivolts. During calibration of one axis, the other axis shows its normal reading.

Footer:

- `1.Exit 2.Back` — During calibration: Btn1 exits calibration, Btn2 goes back
- `1.Done 2.Back` — When complete: Btn1 saves and exits, Btn2 goes back
- `1.Retry 2.Back` — On error: Btn1 restarts calibration, Btn2 goes back
- `1.Continue 2.Cancel` — During adjust max: Btn1 continues, Btn2 cancels

Skip Calibration

Selecting "Skip (Az=El=3.2V)" sets both axes to use fixed voltage ranges (min=0V, max=3.2V) without performing the automatic calibration sweep. Useful when the Yaesu controller's trim pots have already been adjusted to output exactly 3.2V at full scale.

6.7 ELF Elevation Prompt

Appears when ELF mode is toggled (enabled or disabled).

```
Elevate To?
-----------
Horizon (0) [5]
45
90
Skip

. . . . . .
1.Nav  1|Sel  2.Back
```

Option	Effect
Horizon (0)	Sets effective elevation to 0° (default, auto-selected after 5s)
45	Sets effective elevation to 45°
90	Sets effective elevation to 90°
Skip	Leaves elevation unchanged

Controls:

- Btn1 short-press: Cycle through options (resets 5s timeout)
- Btn1 long-press: Select current option
- Btn2: Back (skip, leave elevation unchanged)
- No action for 5 seconds: Auto-selects Horizon (0)

Countdown indicator: `[x]` next to the first option shows remaining seconds before auto-selection. Disappears immediately when any button is pressed.

---

7. Construction Guide

7.1 Parts List

#	Item	Qty	Price (ZAR)	Supplier/SKU
1	BMT ESP8266 NodeMCU WiFi Board [V2-ESP8266 Module ESP-12E NodeMCU Lua (V2) CP2102 USB Interface]	1	R100	Communica SKU: BMT ESP8266 NODEMCU WIFI BOARD
2	HKD ESP8266 NodeMCU Driver Board [WiFi NodeMcu Lua L293D Driver Board for ESP8266 ESP-12E]	1	R69	Communica SKU: HKD ESP8266 NODEMCU DRIVER BOARD
3	BDD I2C 0.96" OLED 128×64-Blue [0.96IN I2C IIC OLED 128×64 Blue. 3.3-5V 160°]	1	R62	Communica SKU: BDD I2C 0.96IN OLED 128X64-BLUE
4	BMT 16-bit A/D (ADC) Dev Board [New ADS1115 16 Bit A/D Module with Internal PGA-I2C. 2-5.5V VCC]	1	R49	Communica SKU: BMT 16 BIT A/D (ADC) DEV BOARD
5	XS82 Straight Inline DIN Plug (Male) • 8 way • Mini	1	R27	Communica SKU: XS82

#	Item	Qty	Price (ZAR)	Supplier/SKU
6	PB11C02 RED Miniature Push Button Switch • Momentary • Form: SPST-0-(1) • 0.3A-50VDC • Solder-Lug • Red-Button • Round Actuator • PTM	2	R20	Communica SKU: PB11C02 RED
7	BT2B NO RIBS [Enclosure ABS 130×70×44 "S20" Black No Ribs]	1	R49	Communica SKU: BT2B NO RIBS
8	BMT Ribbon Jumper 40W F/F 15cm [Jumper Female/Female 15cm in 40Way Colour Cable]	1	R23	Communica SKU: BMT RIBBON JUMPER 40W F/F 15CM
	Subtotal (required parts)		**R399**	
9	PDC-5 Spacer Snap Fit Top Flat Base HT=5mm *(optional, for mounting)*	9	R1.27 each	Communica SKU: PDC-5
10	CAB04PR,22MSCR Mylar Screened Cable 4 pair • 0.22mm² *(optional, for Yaesu DIN connection)*	1 m	R13.80/m	Communica SKU: CAB04PR,22MSCR
	Estimated total (with optional items)		**~R425**	

Notes on optional items:

- **PDC-5 spacers**: These plastic spacers are used to mount the NodeMCU Driver Board and the ADS1115 board inside the enclosure. The plastic posts need to be trimmed slightly to fit through the small mounting holes. They snap through the board on the trimmed side, and the flat base can be glued directly into the enclosure or onto another spacer. This approach lets you lift the internal boards out of the box for servicing without permanently fixing them. Alternative mounting methods (double-sided tape, hot glue, etc.) work equally well.
- **CAB04PR,22MSCR cable**: This 4-pair screened cable is suitable for the 8-core Yaesu DIN-8 connection. Any quality 8-core screened cable with appropriate gauge will work.

7.2 Tools Required

- Drill and drill bits (various sizes)
- File (for smoothing drilled holes)
- Soldering iron and solder
- Wire strippers
- Screwdriver (for terminal blocks)
- Multimeter (for voltage verification)

7.3 Box Preparation

1. Select a plastic enclosure large enough to hold the motor shield with the ESP8266 plugged in, plus the OLED display.
2. Mark a rectangle on one face of the enclosure for the OLED display, ensuring it will be visible from the front.
3. Drill out the marked rectangle and file the edges smooth.
4. Drill two holes for the push buttons.
5. Drill one hole for the 8-core screened cable (Yaesu DIN connection).
6. File a small slit near the top of the lid for the micro-USB cable, or drill a hole and cut/resolder the USB plug.

7.4 Mounting the OLED

1. Insert the OLED display from inside the enclosure so the screen faces outward through the cutout.
2. **Critical orientation**: The OLED must be mounted with the pin headers on the top edge. If you mount it upside down (pins on bottom), the text will display upside down on the screen.
3. Secure internally using insulation tape and/or black duct tape.
4. Ensure the display is firmly held and the screen is flush with the enclosure face.

7.5 Using the Ribbon Jumper Cable

The 40-way female-to-female ribbon jumper cable is purchased as a convenient source of coloured jumper wires — **it is not used as a single ribbon cable**. You will strip it apart and use the individual wires for various connections throughout the build.

How to use the ribbon jumper cable:

1. **Separate individual wires**: Pull apart the ribbon cable to get individual female-to-female jumper wires. Each wire has a female header connector on both ends and is a different colour, making identification easy.
2. **Keep wires bundled where possible**: When connecting multiple wires to the same destination (e.g., several wires going to the motor shield's GPIO headers), keep them bundled together as a small group of 2–4 wires. This makes the wiring neater and easier to trace.
3. **Solder where practical**: For connections to the ADS1115 ADC board, **cut off the female header connector on one end and solder the bare wire directly to the board's pads**. The ADS1115 module typically does not come with pin headers pre-soldered — this is intentional and convenient. Direct soldering is much more reliable than using jumper connectors, which can wobble loose over time due to vibration from the rotator motors.
4. **I2C splitter cables**: The I2C bus requires a "Y-splitter" to connect both the OLED display and the ADS1115 to the same SDA and SCL lines. Make these splitter cables from the ribbon jumper wires:
   - Take one jumper wire, cut it in the middle
   - Strip the insulation from both cut ends
   - Twist each cut end around a third jumper wire (the "branch")
   - Solder each joint and insulate with heat shrink or insulation tape
   - Repeat for both SDA and SCL lines
5. **Motor shield connections**: For wires going to the motor shield's GPIO pin headers (D1–D8), use the female header connector on one end of the jumper wire plugged into the shield. On the other end:
   - If connecting to a button: cut off the female connector and solder the bare wire to the button terminal
   - If connecting to the ADS1115: cut off the female connector and solder the bare wire to the board's pads

Summary: Think of the ribbon jumper cable as a convenient source of pre-coloured, pre-tinned jumper wires with connectors already crimped on one end. You'll use the connector end where it plugs into a header (like the motor shield's GPIO pins), and cut and solder the other end wherever a more permanent, reliable connection is needed.

7.6 Wiring the Buttons

Btn1 (D8 / GPIO15):

- Connect one terminal to D8 on the motor shield.
- Connect the other terminal to any 3.3V rail pin on the motor shield.
- This button is active HIGH — when pressed, D8 goes from 0V to 3.3V.

Btn2 (D5 / GPIO14):

- Connect one terminal to D5 on the motor shield.
- Connect the other terminal to any GND rail pin on the motor shield.
- This button is active LOW — when pressed, D5 goes from 3.3V to 0V.

Why the difference? GPIO15 must be LOW at reset for the ESP8266 to boot normally. The motor shield comes with a fixed pull-down resistor on this pin. If you wired Btn1 to GND instead of 3.3V, pressing it would hold GPIO15 LOW and prevent booting. GPIO14 has a programmable pull-up resistor used in firmware.

7.7 Wiring the I2C Bus

How I2C works (brief explanation): I2C is a communication method that uses just two wires — SDA (data) and SCL (clock). Multiple devices can share these same two wires simultaneously. Each device has a unique address, so when the ESP8266 wants to talk to one specific device, it sends that device's address first. The other devices on the bus simply ignore the message. This is why it's perfectly fine to connect both the OLED and the ADS1115 "in parallel" on the same two wires — they won't interfere with each other because they have different addresses (OLED = 0x3C, ADS1115 = 0x48).

OLED Display (pins from left to right when viewing from front):

- Pin 1 (GND) → Any GND rail pin
- Pin 2 (VCC) → Any 3.3V rail pin
- Pin 3 (SCL) → D7 (GPIO13) on motor shield, using one end of a 3-leg splitter
- Pin 4 (SDA) → D6 (GPIO12) on motor shield, using one end of a 3-leg splitter

ADS1115 Module:

- GND → Same GND rail as OLED (or another GND pin)
- VCC → Same 3.3V rail as OLED (or another 3.3V pin)
- SCL → Third leg of the SCL splitter wire from OLED
- SDA → Third leg of the SDA splitter wire from OLED

Solder the connections to the ADS1115 module for mechanical security rather than using jumper blocks.

Heat-shrink recommendation: When making the I2C splitter cables, slide a small piece of heat-shrink tubing onto each wire before soldering. After soldering the bare wire ends to the OLED's header pins, slide the heat-shrink over the joint and shrink it in place with a heat gun or lighter. This prevents accidental shorts between adjacent solder joints. When cutting and stripping the jumper wires for the splitters, leave enough extra length so you have room to fit the heat-shrink tubing over the solder joints.

7.8 Wiring the Yaesu DIN-8 Connector

Map out the wires within the 8-core screened cable connected to the DIN plug. Connect as follows:

Yaesu DIN Pin	Function	Motor Shield Terminal
Pin 1	Elevation feedback voltage	ADS1115 A1
Pin 2	Azimuth CW	A+
Pin 3	Elevation Down	B+
Pin 4	Azimuth CCW	A-
Pin 5	Elevation Up	B-
Pin 6	Azimuth feedback voltage	ADS1115 A0
Pin 7	+12V from Yaesu controller	VM
Pin 8	Common ground	GND

Practical tips:

- **Do this wiring before installing the motor shield in the enclosure**. With the shield sitting on your workbench, it's much easier to position the wire ends into the screw-down terminals. If your enclosure is tight-fitting, trying to manoeuvre wires into terminal positions from inside the box can be very difficult.
- **Keep the DIN plug disconnected from the Yaesu controller** while making these connections. Once all wires are connected and terminals tightened, use a multimeter to verify each wire goes to the correct DIN pin before plugging anything in.
- Tighten all terminal screws securely.

7.9 Power Connections

VIN (Logic Power):

- Powered via USB (5V) from a computer, USB charger, or external 5V supply.
- The on-board voltage regulator can handle up to 10V. Higher voltages will damage the unit.
- The module can draw up to ~400mA during WiFi transmission.

VM (Motor Power):

- Supplied by the Yaesu controller's +12V output (Pin 7 of DIN-8).
- Used by the motor shield to drive the Yaesu relay inputs.

CRITICAL WARNING:

- **Do NOT install the jumper that shorts VIN and VM** on the motor shield.
- VIN runs at 5V (from USB), VM runs at 12V (from Yaesu controller).
- Shorting them would feed 12V into the USB socket and destroy the ESP8266 module.

7.10 Potentiometer Calibration (Before First Power-Up)

IMPORTANT: The Yaesu controller outputs ~4.5V by default. The ADS1115 can only handle up to 3.3V safely. You MUST adjust the trim pots on the Yaesu controller before connecting position feedback voltages to the ADS1115.

Prerequisite: The Yaesu controller must already be wired to the external azimuth/elevation rotator. The trim pot calibration is done with the actual rotator cable installed, because cable length affects voltage drop.

CRITICAL WARNING: If you change the length of the cable between the Yaesu controller and the rotator, you MUST redo this trim pot calibration. Shortening the cable reduces voltage drop, which means the voltage at the Yaesu controller's feedback pins will be higher than what you calibrated. If this voltage exceeds 3.3V, it will destroy the ADS1115 module and it will have to be replaced. **Always recalibrate after any cable length change.**

Steps:

1. Rotate the azimuth rotator clockwise to the limit (maximum meter deflection).
2. Locate the two trim pots on the Yaesu controller (adjacent to the meter calibration pots).
3. Measure the voltage between Pin 6 and Ground at the DIN connector.
4. Adjust the **Azimuth** trim pot until you read exactly 3.2V.
5. Rotate the azimuth rotator counter-clockwise to the limit. Verify the voltage approaches 0V.
6. Rotate the elevation rotator UP to the limit (maximum meter deflection).
7. Measure the voltage between Pin 1 and Ground at the DIN connector.
8. Adjust the **Elevation** trim pot until you read exactly 3.2V.
9. Rotate the elevation rotator DOWN to the limit. Verify the voltage approaches 0V.

Only after both pots are calibrated should you connect the feedback voltages to the ADS1115 and power up AZELCtrl.

7.11 Firmware Installation

The ESP8266 must first be flashed with the AZELCtrl firmware. You can either build it from source (Arduino IDE / `arduino-cli`) or flash the pre-built firmware binary from the release package — the web-tool method below works identically on Windows and macOS. Flashing is platform-specific; follow the relevant subsection.

Where to get the files: the current release is always staged in the `release/` folder at the root of the repository — that single folder holds all four files you need (firmware + the three SPIFFS uploads). Download them individually from there, or grab the bundled `azelctrl-v1.5.x.zip` from the repo root. The folder is refreshed by the release builder on every update, so it always matches the latest firmware.

Download the **release package** (containing four files):

File	Purpose
`azelctrl-firmware.bin`	The compiled firmware — this is the main program
`index.html.gz`	The satellite-tracker web interface (gzip-compressed)
`favicon.ico`	Browser tab icon
`world_coastline.js.gz`	World-map coastline data for the orbital map (gzip-compressed)

> Only `azelctrl-firmware.bin` is flashed to the chip. The other three files are uploaded to the ESP's filesystem (SPIFFS) afterwards via the web interface — see Section 8.3.

7.11.1 Windows

1. Install the USB driver

Connect the ESP8266 via USB and install the driver for its USB-to-serial chip:

Board chip	Driver
CH340G (most NodeMCU clones)	CH340 driver (WCH)
CP2102 (NodeMCU v2, Wemos D1 Mini)	CP210x driver (Silicon Labs)

Open **Device Manager** → **Ports (COM & LPT)** and note the COM port (e.g. `COM3`). A device with a yellow triangle means the driver is missing — install it and reconnect. Try a different USB cable if the board is never detected (charge-only cables carry no data).

2. Flash using ESPHome Web (recommended)

The easiest method — a browser-based tool using Web Serial; no extra software needed.

1. Open https://web.esphome.io in **Chrome, Edge, or Firefox 151+** (Safari does not support Web Serial).
2. Click **CONNECT** and select the ESP8266's COM port.
3. Click **INSTALL** (not "Prepare for first use"), then **CHOOSE FILE** and select `azelctrl-firmware.bin`.
4. Flash — the ESP reboots automatically when done.

If you prefer a desktop tool, the **Arduino IDE** also works: add the ESP8266 boards URL `https://arduino.esp8266.com/stable/package_esp8266com_index.json`, install **esp8266 by ESP8266 Community** (v3.1.2+), then select board **NodeMCU 1.0 (ESP-12E Module)** with **Flash Size = 4MB (FS:3MB OTA:~512KB)**, Flash Mode **DIO**, CPU **80 MHz**, Upload Speed **115200**, and the COM port before **Sketch** → **Upload** (if it hangs at `Connecting...`, hold **FLASH**, tap **RST**, release **FLASH**, retry). The 4MB/FS:3MB layout is **critical** — without it the filesystem can't hold the web interface and map data.

7.11.2 macOS

1. Install the USB driver

CP2102-based boards (like the NodeMCU v2 in the parts list) need the CP210x VCP driver (Silicon Labs); CH340-based boards need the CH340 macOS driver (WCH). After installing, confirm the board appears in **System Information** → **USB** (approve the extension in **System Settings** → **Privacy & Security** if prompted).

2. Identify the serial port

Plug in the board and run:

```
ls /dev/cu.*
```

The board appears as `/dev/cu.usbserial-xxxx` (e.g. `/dev/cu.usbserial-0001`). Note this path — you'll need it below.

3. Flash using ESPHome Web (recommended)

Same tool as Windows — Web Serial works in **Chrome or Edge on macOS** (not Safari):

1. Open https://web.esphome.io in Chrome/Edge.
2. Click **CONNECT**, select the ESP8266's serial port.
3. Click **INSTALL** → **CHOOSE FILE** → select `azelctrl-firmware.bin`.
4. Flash — the ESP reboots when done.

4. Alternative: build & upload with `arduino-cli` (terminal)

If you build from source (e.g. the `ycint/` firmware sketch):

```
# One-time setup
arduino-cli config init
arduino-cli core update-index
arduino-cli core install esp8266:esp8266

# Compile (4M3M layout — 4 MB flash, 3 MB filesystem)
arduino-cli compile --fqbn esp8266:esp8266:nodemcuv2:eesz=4M3M ycint/

# Upload firmware to the board
arduino-cli upload --port /dev/cu.usbserial-0001 \
  --fqbn esp8266:esp8266:nodemcuv2:eesz=4M3M ycint/
```

The Arduino IDE (macOS app) with the same ESP8266 board package and board/flash settings is an equivalent option.

After flashing (both platforms):

1. Power up and confirm the OLED shows the startup sequence ("AZELCtrl", firmware version).
2. **First boot with no saved settings:** the firmware detects the incomplete settings, shows **"Missing Settings!"**, and **automatically starts in Access Point mode** — SSID **AZELCTRL_XXXX** (password **yaesu123**). Connect to that network and open **http://192.168.4.1/** in a browser. The web interface is served immediately — no OLED menu navigation is required first.
3. On the **Settings** tab you can do both of the following:
   - **Upload the supporting files** — `index.html.gz` via **Upload Web Interface**, then `favicon.ico` and `world_coastline.js.gz` via **Upload Any File**.
   - **Enter your home WiFi details** — type your home network's name and password into **Station SSID** and **Password**, click **Save**, then set **WiFi Mode** to **Station** and click **Apply**. The unit reboots and joins your home network (look up its DHCP-assigned IP in your router's client list; reconnect this device to your home WiFi first). You can always switch back to AP mode later from the WiFi settings on this page or the OLED Network menu.
4. Refresh the web page — the tracker UI should load and the **Map** tab should show coastlines.

Note: WiFi credentials are not stored in the source code — they are entered on a live unit, either on the web interface's **Settings** page (Station SSID / Password) or via the captive portal setup process (see section 7.12), and saved to EEPROM.

7.11.3 Upgrading a Unit That Is Already Configured

If your unit is already set up and working (WiFi connected, calibrated, satellites configured), updating it takes three steps — **no reconfiguration of any kind is required**:

1. **Flash the new firmware.** Connect the unit to your computer over USB and flash `azelctrl-firmware.bin` with web.esphome.io exactly as described above (INSTALL → CHOOSE FILE). The unit reboots automatically.
2. **Reconnect over the network.** All your settings — WiFi credentials, rotator model, calibration, protocol, display options — live in the unit's EEPROM and are **not touched by a firmware flash**. The unit simply rejoins your home network as before; if its IP changed, look it up in your router's client list.

3. **Re-upload the three supporting files.** Open the web interface, go to **Settings** → **ESP Management**, upload `index.html.gz` (Upload Web Interface), then `favicon.ico` and `world_coastline.js.gz` (Upload Any File). Refresh the page — the tracker should load and the **Map** tab should show coastlines.

Two things worth knowing:

- **You do not need to "reconfigure" SPIFFS.** The size and location of the filesystem are baked into the firmware binary at build time — the release binary uses the 4 MB / 3 MB filesystem layout. Flashing `azelctrl-firmware.bin` switches the chip to that layout automatically; no tool options or settings are involved. If your unit was flashed with older firmware that used a smaller filesystem, its old contents simply become unused space, which is why step 3 is mandatory after such an upgrade. Between recent releases (same layout) the files survive the flash, but re-uploading all three is still recommended so you definitely have the current versions.
- **Check the file list after uploading.** The **ESP Management** file list should show exactly three files (`index.html.gz`, `world_coastline.js.gz`, `favicon.ico`). In rare cases — when a previous firmware version used a different flash layout — leftover copies from the older filesystem can reappear in the list; delete any extras so only the current three remain.
- **Re-uploading the files is required after every firmware flash that changes the filesystem layout**, and recommended after every other flash. It takes under a minute and is the only step users commonly miss after an update.

7.12 Configuring WiFi via Captive Portal

When AZELCtrl has no WiFi credentials saved (fresh flash or after wiping all settings), you can configure WiFi using the captive portal:

1. From the SETUP menu, select "Network" → "WiFi Mode(None)".
2. AZELCtrl starts an open WiFi access point named **AZELCTRL_XXXX** (where XXXX is the last 4 digits of the device MAC address).
3. The OLED shows: "Connect phone to: AZELCTRL_XXXX" and "IP: 192.168.4.1".
4. On your phone or laptop, open WiFi settings and connect to the **AZELCTRL_XXXX** network (no password required).
5. Open a web browser and navigate to any address (e.g., 192.168.4.1). You'll see the AZELCTRL Setup page.
6. Enter your home WiFi SSID (network name) and password.
7. Tap "Save & Connect". AZELCtrl saves the credentials to EEPROM and stops the captive portal.
8. Disconnect your phone/laptop from the AZELCTRL_XXXX network.
9. On the AZELCtrl OLED, press Btn1 to return to the Network menu, then select "Connect WiFi" and press Btn2 to start connecting to your home network.
10. Wait for the connection to complete. If successful, the IP address will be displayed.

This same process can be used at any time to change WiFi credentials — simply go back to SETUP → Network → Set WiFi Creds.

---

8. Operating the Device

8.1 Power-Up Sequence

1. Connect power via USB (or VIN terminal).
2. OLED displays the splash screen for 3 seconds:
   - Product name, firmware version, date, author
3. If mandatory settings are incomplete (rotator variant, calibration, WiFi mode):
   - A "Missing Settings!" notice appears for 3 seconds
   - The device automatically starts WiFi in **Access Point (AP) mode** — look for the **AZELCTRL_XXXX** network (password: **yaesu123**)
   - The web interface is available at **http://192.168.4.1/** for immediate configuration
   - The OLED then shows the SETUP menu for completing the mandatory settings
4. Otherwise, the Boot menu appears:
   - "Rotator Ctrl" with a 3-second countdown (auto-selects if no button pressed)
   - "Setup" option
5. If "Rotator Ctrl" is selected and WiFi is configured:
   - WiFi connecting screen appears with SSID and elapsed time
   - On success: Connected screen with IP address, signal strength, 5-second countdown (press any button to skip)
   - On failure: "WiFi Failed!" screen for 2 seconds
6. OLED switches to the Az/El control screen.

8.2 Manual Control via Buttons

From the main Az/El control screen:

1. Press **Btn1** to cycle through options: SETUP → LEFT → RIGHT → UP → DOWN → ELF
2. Press and **hold Btn2** to move in the selected direction (LEFT/RIGHT/UP/DOWN). Release to stop.
3. Selecting SETUP and pressing Btn2 opens the SETUP menu.
4. Selecting ELF and pressing Btn2 toggles ELF (over-the-top) mode and shows the "Elevate To?" prompt.
5. When SETUP is selected, holding both Btn1+Btn2 together reboots the device.

8.3 Remote Control via Web Interface

The web interface provides a comprehensive browser-based control panel for operating the rotator from a phone, tablet, or computer. All orbital computation runs in your browser — no external servers required.

Accessing the web interface:

- **AP Mode**: On first boot (or after factory reset), the ESP automatically starts in AP mode. Connect to the **AZELCTRL_XXXX** WiFi network (password: **yaesu123**), then open a browser to **http://192.168.4.1/**.
- **Station Mode**: Connect to the same WiFi network as AZELCtrl, then open a browser to the IP address shown on the OLED. The IP is often 192.168.x.x, or 10.0.x.x on the default home network.

The web interface is fully functional even while navigating the device's OLED menus — both work independently.

Web interface tabs:

Tab	Purpose
Map	Interactive world map showing satellite position, ground track, footprint, terminator, Sun/Moon
Tracking	Real-time satellite tracking with polar plot, AOS/LOS timer, frequency display
Satellites	Browse and manage satellite passes and frequencies
Manage	Add, remove, and refresh TLE data for satellites
Passes	Predict upcoming passes for the selected satellite
Control	Manual rotator control and calibration
Settings	Station location, tracking preferences, device configuration, ESP management

Header Bar

The header bar is visible on all screens.

Element	Description
AZELCtrl Satellite Tracker	Application title
Satellite dropdown (▸)	Select any loaded satellite, the Sun, or the Moon. Use ↑/↓ arrow keys to cycle.
TLE age (e.g. "2h ago")	How old the selected satellite's TLE data is. Green (<24h), orange (24–72h), red (>72h).
Connected / Disconnected	Green dot = rotator reachable. Red = no connection.

Map Tab

Displays an offline, equirectangular-projection world map rendered on an HTML Canvas — no external mapping API required.

Map Overlays:

Toggle	Default	Description
Day/Night	Off	Shows the day/night terminator (sunrise/sunset lines and shaded night side)
Footprint	On	Green translucent circle showing the satellite's area of visibility
Grid	On	Latitude/longitude grid lines every 30° lat, 60° lon
🌙 **Moon**	Off	Shows the Moon's current sub-point
☀️ **Sun**	Off	Shows the Sun's current sub-point

Map Controls:

Drag to pan, scroll/pinch to zoom. The **Duration** selector sets the ground-track length (30 min to 6 hours).

Control	Description
↺ **Reset View**	Re-centres the map on your station (QTH) and resets zoom to 100%
⊕ **Center Sat**	Re-centres the map on the satellite's current ground position
◎ **Follow Sat / Stop Following Sat**	Toggle that keeps the map centred on the satellite on every position update (30 s) — press again to stop and release the view

Control	Description
ZOOM: 1 2 3 4	Centred preset zoom buttons: 1 = 100% (full globe), 2 = 75%, 3 = 50%, 4 = 25%. Zooms anchor on your station, or on the satellite's current position while Follow Sat is active
Map footer	Colour-coded live values: **POS** (blue), **ALT** (amber), **VEL** (purple), **DIST** (green)

Vertical panning is clamped so the view never scrolls past ±90° latitude (the poles); at 100% the map always shows the full pole-to-pole globe. The map refreshes the satellite position every 30 seconds.

Tracking Tab

The primary tab for operating the rotator.

Satellite Info shows current Azimuth and Elevation of the target, with AOS/LOS timer and range.

Rotator Info shows current rotator Azimuth and Elevation positions with movement arrows.

ELF Mode Toggle: Enable Elevation Flip Mode for passes that cross the 180° wrap boundary.

Polar Plot: Circular plot showing satellite trajectory (path traveled in solid blue, future path in dashed green/orange), current satellite position (blue diamond), and rotator position (green circle).

Frequency Display (satellite mode): Shows downlink/uplink frequencies with Doppler correction when SatNOGS data is available. The frequency table includes the following columns:

Column	Description
DL Freq MHz	Downlink frequency range (lower–upper)
RX Doppler	Doppler-corrected receive frequency (green = positive, red = negative)
UL Freq MHz	Uplink frequency range (lower–upper)
TX Comp	Doppler compensation for uplink (opposite sign of RX Doppler)
Mode	Operating mode (e.g., FM, USB, LSB, CW, etc.)
CTCSS	Continuous Tone-Coded Squelch System tone (editable, in Hz)
Wake Tone	Dual-tone CTCSS wake tone for FM repeaters (editable, in Hz)

Conditional tone column visibility: The CTCSS and Wake Tone columns only appear when at least one visible transmitter has **FM mode** with an **uplink frequency**. For individual rows:

- If the mode is FM **and** an uplink frequency exists → editable input fields are shown
- If the mode is not FM (USB, LSB, CW, etc.) **or** no uplink frequency is present → `–` is displayed (no tone fields)

Editing tone values: Click into any CTCSS or Wake Tone input field to edit. Changes are saved automatically when the field loses focus (blur) or when you press Enter. Press Escape to revert to the last saved value. User-edited values are stored in your browser's local storage and take precedence over values fetched from the SatNOGS database during frequency refreshes. This allows you to correct inaccurate CTCSS data or add tones that SatNOGS does not provide.

Dual-tone CTCSS: Some FM repeater satellites require two tones — a **Wake Tone** to activate the transponder, and a **CTCSS** tone for normal operation. Both fields are editable independently and are only displayed for FM transmitters with an uplink frequency.

Satellites Tab

Displays all loaded satellites with upcoming pass information. Each row shows satellite name, AOS time, LOS time, AOS azimuth, max elevation, duration, and countdown timer. Filters allow showing only active satellites, frequencies, and hiding individual satellites. When the **Show Frequencies** filter is enabled, each satellite's pass table expands to show downlink/uplink frequencies, mode, and CTCSS tone (displayed as `–` for non-FM modes or when no uplink frequency is present).

Manage Tab

This tab is where you manage your satellite catalogue — adding satellites, downloading TLE orbital data, and fetching frequency information.

Manage Added Satellites: Shows a compact list of all satellites you have added, with TLE age indicators (time since last TLE update) and remove buttons.

Adding Satellites:

Three methods are available via the sub-tabs:

CelesTrak Add — Browse and add satellites from the CelesTrak catalogue:

1. Click **Load List from CelesTrak** to fetch the current amateur radio satellite catalogue.
2. Browse the list, filtered by frequency band if desired.
3. Click **Add** next to any satellite to import it with fresh TLE data.
4. Use **Add ALL** to import every satellite in the current view at once.
5. Click **Refresh All TLEs** at any time to update orbital data for all loaded satellites.

Paste Add — Add a satellite manually:

1. Enter the satellite name and NORAD ID.
2. Paste the two-line TLE (from CelesTrak, Space-Track, or another source) or just the NORAD ID.
3. Click **Add** to parse and store the satellite. TLE data will be fetched automatically if only a NORAD ID was provided.

SatNOGS Data — Fetch frequencies and status from the SatNOGS database:

The SatNOGS Data panel provides two data sources. Read the CORS note below first — it explains which options will work with your setup.

Live API (direct):

1. Select a satellite from the dropdown (or leave **All satellites** selected).
2. Click **Refresh Status** to fetch active/inactive status from db.satnogs.org.
3. Click **Refresh Frequencies** to download mode, frequency, and Doppler correction data.
4. These fetch from the live SatNOGS API, optionally proxied through the ESP.

Alt Sync (pre-cached CSV): Located below the live API section, these buttons fetch from pre-generated CSV files hosted online. This bypasses the CORS/proxy problems entirely and is much faster. Data may be less current than the live API.

1. **Alt Sync Status** — Downloads a pre-generated CSV file of satellite statuses (active/inactive/decayed) and updates all matching satellites in your catalogue.
2. **Alt Sync Frequencies** — Downloads a pre-generated CSV file of satellite frequencies and modes, updating all matching satellites.
3. Both methods display a status line showing how many satellites were updated and when the data was generated.

CORS and Proxy Note:

SatNOGS's API does not currently serve CORS headers, which means browsers will block JavaScript requests made directly to `db.satnogs.org`. The Settings tab provides three proxy modes to work around this:

Mode	Description	Best For
Use ESP proxy	Requests go through the ESP8266, which fetches from SatNOGS and returns the response. The ESP sets CORS headers so the browser accepts it.	Occasional small queries
Use custom proxy URL	You provide a separate CORS-enabled proxy server (e.g. a Python script on your PC, or a cloud function). Enter its URL in the field.	Reliable, large-scale fetching
Direct (no proxy)	Browser fetches directly from SatNOGS. **Will not work** until SatNOGS adds CORS support.	—

Limitations of the ESP proxy:

- **Slow**: The ESP8266 must establish an HTTPS connection for each request. This takes approximately 30 seconds per request. Refreshing status + frequencies for 20 satellites can take over 20 minutes.
- **Limited RAM**: The ESP8266 has limited heap memory. Responses larger than approximately 4 KB will be rejected with a "Response too large" error. Some satellites in the SatNOGS database have dozens of frequency entries that together exceed this limit.
- **Recommendation**: For reliable operation with a full satellite catalogue, use a **Custom proxy URL** (a lightweight CORS proxy running on your PC or LAN) or use **Alt Sync** for the pre-cached data.

SatNOGS Proxy settings are located in the **Settings** tab under the **SatNOGS Proxy** card.

Data from both sources is cached in your browser and used to display frequency tables with Doppler-shift-compensated uplink/downlink frequencies on the Tracking tab.

TLE and Frequency Data Storage: All satellite data (TLEs, frequencies, status) is stored in your browser's local storage. This means:

- Data persists across browser sessions on the same device.
- Clearing your browser data will remove your satellite list.
- Each device/browser combination has its own satellite list — there is no server-side storage.

Passes Tab

Shows detailed pass predictions for the selected satellite, Sun, or Moon. Select lookahead window (24/48/72 hours). Lists each pass with AOS time, LOS time, AOS azimuth, max elevation, and duration.

Control Tab

Shows current rotator position. Provides a cross-shaped control pad with UP, DOWN, LEFT, RIGHT, and STOP buttons (press and hold to move, release to stop). Manual position entry allows setting precise azimuth/elevation

values. Start Calibration button initiates the potentiometer calibration sweep (same as OLED menu).

Settings Tab

Ground Station Location: Enter Maidenhead grid square or lat/lon/altitude.

Tracking Preferences: Update interval, background keep-alive, decimal degrees, minimum elevation, azimuth/elevation speed, lead angle, dead zone.

Decimal Degrees (off by default): When off, azimuth and elevation are shown as whole degrees (rounded) on both the OLED display and the web interface. When enabled, they show one decimal digit (e.g. `237.5` instead of `238`). The setting is stored on the device itself, so it survives power cycles and applies to every client — the web page syncs the toggle from the device when it loads. It only affects how values are *displayed*: all tracking, prediction, and command calculations continue at full internal precision regardless of the setting.

Rotor Power Detection (on by default): Shows whether the Yaesu controller is connected and powered (see §3.3 for the optional divider wiring). When the controller is unplugged or unpowered, the Tracking tab shows a red **ROTOR OFFLINE** banner, the OLED replaces its selection line with a persistent `ROTOR OFFLINE` indicator, and each transition is logged to the protocol debug log. Disable it while the detection resistors are not yet fitted, to suppress the warning. Stored on the device; survives power cycles.

Background Keep-Alive (on by default): Browsers throttle — and after a few minutes fully freeze — JavaScript in tabs that are not in focus, which can stop tracking while you work in another tab. When enabled, AZELCtrl plays an inaudible audio stream (~–78 dB, effectively silent) during tracking so the browser keeps the tab awake, and requests a screen wake lock so phones do not suspend the page when the screen turns off. If the tab was frozen anyway, tracking resynchronises automatically the moment you switch back to the tab. This has no effect on the AZELCtrl device itself.

Controller: | Field | Description | |-------|-------------| | **Rotator Model** | Select N360, S360, N450, or S450. Click **Set** to apply. Changing model clears calibration. | | **Calibration** | Displays status. **Load from ESP** refreshes values. **Skip Calibration** sets default 0–3.2V range. Manual voltage entry for Az/El Min/Max. |

WiFi: | Field | Description | |-------|-------------| | **WiFi Mode** | Select **Access Point**, **Station**, or **Disabled**. Click **Apply** to switch (disconnect warning shown). | | **Station SSID / Password** | Enter home WiFi credentials and click **Save**. | | **AP Password** | Change the 8-character AP password. | | **Current AP SSID** | Read-only display of the ESP's AP network name. |

PC Interface: | Field | Description | |-------|-------------| | **Protocol Variant** | Select GS-232A, GS-232B, or Easycom. Click **Set** to apply. | | **Baud Rate** | Select serial baud rate (9600–115200). | | **Show Commands** | Toggle display of received commands on the OLED. | | **Reverse Delay** | Seconds to wait before reversing direction (0–3s). | | **Debug USB Log** | Opens a live log of serial/TCP protocol traffic in a new browser tab — for diagnosing external tracking software. See Debugging External Tracking Software. |

Network: | Field | Description | |-------|-------------| | **Hamlib TCP Server** | Toggle the rotctld-compatible server on port 4533. |

ESP Management: Upload web interface, upload any file, list/delete files on the ESP filesystem.

Debug: JavaScript logging toggle, export/load configuration JSON.

Keyboard Shortcuts

Shortcut	Action
↑/↓ **arrows** (dropdown focused)	Cycle through satellites
Ctrl+Shift+R	Hard refresh browser cache
F12	Open browser developer console

8.4 Connecting Tracking Software

AZELCtrl supports two distinct methods of satellite tracking:

Method 1 — Browser-based Tracking (Built-In): AZELCtrl's web interface includes a full satellite tracking engine that runs entirely in your browser using the satellite.js library. All orbital computations (position prediction, AOS/LOS timing, Doppler shift, polar plots, world map) happen in the browser — no external software required. Simply:

1. Connect to the web interface (via AP or Station mode — see section 8.3).
2. Go to the **Manage** tab to add satellites and download TLE/frequency data (see Manage Tab in section 8.3).
3. Select a target satellite from the header dropdown.
4. Go to the **Tracking** tab and click **Start** to begin automated tracking.

This is the simplest method and works on any device with a browser — phone, tablet, laptop, or desktop.

Method 2 — External Tracking Software: AZELCtrl can also be controlled from dedicated satellite tracking software (Gpredict, Rotor, etc.) via USB serial or TCP/IP. Choose whichever interface suits your setup. This method gives you access to advanced features like predictive pass planning and multi-satellite scheduling that dedicated desktop software provides.

The sections below describe the connection details for external software.

USB Serial Connection

1. Connect AZELCtrl to your computer via USB cable.
2. Your operating system will recognize it as a virtual COM port:
   - **Windows**: Check Device Manager → Ports (COM & LPT) — typically COM3 or higher
   - **macOS**: /dev/cu.usbserial-* (check System Information → USB)
   - **Linux**: /dev/ttyUSB0 or /dev/ttyUSB1
3. Ensure AZELCtrl's **Hamlib Server** setting is **No** (so it speaks GS-232/Easycom over serial).
4. In your tracking software, configure a new rotator connection:
   - Type: GS-232B (or your configured protocol)
   - Connection: Serial / COM port
   - Port: Your virtual COM port (e.g., COM3)
   - Baud rate: 115200
5. Test the connection.

TCP/IP Network Connection

1. Ensure AZELCtrl is connected to your WiFi network (check OLED for IP address).

2. In your tracking software, configure a new rotator connection:
    - Type: Depends on Hamlib Server setting (see below)
    - Connection: TCP/IP
    - Host: AZELCtrl's IP address (shown on OLED)
    - Port: 4533
3. Match the Hamlib Server setting to your software:
    - **Hamlib Server = Yes**: Software expects rotctl text protocol (Gpredict)
    - **Hamlib Server = No**: Software expects GS-232/Easycom (Rotor, others)

Gpredict

Gpredict supports both USB serial and TCP/IP connections.

Via TCP/IP (recommended):

1. Open Gpredict preferences → Satellite → Control → Rotator control.
2. Set type to "Hamlib rotctld".
3. Set host to AZELCtrl's IP address.
4. Set port to 4533.
5. **Critical**: Ensure AZELCtrl's **Hamlib Server** setting is **Yes** (under Network in the SETUP menu). Without this, AZELCtrl will speak GS-232/Easycom over TCP instead of the rotctl protocol that Gpredict expects, and the connection will fail.
6. Enable tracking for the desired satellite.

Via USB Serial (using rotctld as bridge): Gpredict does not connect directly to serial ports. It requires the Hamlib rotctld daemon as an intermediary.

1. Connect AZELCtrl to your computer via USB.
2. Ensure AZELCtrl's **Hamlib Server** setting is **No** (so it speaks GS-232/Easycom over serial).
3. Start rotctld, pointing it at AZELCtrl's virtual COM port with the matching protocol (e.g., GS-232B at 115200 baud).
4. In Gpredict preferences → Satellite → Control → Rotator control:
    - Set type to "Hamlib rotctld"
    - Set host to localhost (or wherever rotctld is running)
    - Set port to the port rotctld is listening on (typically 4532)
5. Enable tracking.

The connection path is: Gpredict → rotctld → AZELCtrl (USB serial).

Note on ELF (over-the-top) mode: Gpredict implements the elevation-flip technique natively. When a satellite pass crosses the rotator's wrap-around boundary, Gpredict automatically transforms the coordinates and sends the appropriate physical targets. **Do not enable ELF mode on AZELCtrl when using Gpredict** — doing so would apply the transformation twice, resulting in incorrect antenna pointing.

Rotor

Rotor supports both USB serial and TCP/IP connections.

macOS users: Rotor works well with AZELCtrl via USB serial. However, Rotor's TCP/IP interface currently does not support elevation control, making it unsuitable for satellite tracking with an azimuth-elevation rotator. Use USB serial or Gpredict for full Az/El control.

Via TCP/IP:

1. Configure a new rotator connection.
2. Set type to GS-232B (or your configured protocol).
3. Set connection to TCP/IP.
4. Set host to AZELCtrl's IP address.
5. Set port to 4533.
6. Ensure AZELCtrl's Hamlib Server setting is **No**.
7. Test the connection.

Via USB Serial:

1. Connect AZELCtrl to your computer via USB.
2. Configure a new rotator connection.
3. Set type to GS-232B (or your configured protocol).
4. Set connection to Serial.
5. Select the virtual COM port for AZELCtrl.
6. Set baud rate to 115200.
7. Ensure AZELCtrl's Hamlib Server setting is **No**.
8. Test the connection.

Other Tracking Software

Many satellite tracking programs support serial connections but not TCP/IP. AZELCtrl's USB serial interface works with any software that supports GS-232A, GS-232B, or Easycom protocols. Simply connect via USB and configure the virtual COM port as you would for any serial rotator controller. Replies are always sent back over the same interface the command arrived on — a serial command is answered on the serial wire, a TCP command over TCP — so the two interfaces never interfere with each other even when both are in use at once.

Debugging External Tracking Software

If your tracking software behaves oddly (timeouts, repeated retries, the antenna not moving as expected), AZELCtrl can capture the exact protocol exchange so the problem can be pinned down — on the software side or the device side.

1. Go to **Settings** → **PC Interface** → **Debug USB Log** and click **Open Debug Tab**.
2. A new browser tab opens showing a live log of everything that crosses the protocol interfaces, exactly as seen by the device:
   - `RX SER` — a command line received on the USB serial port (control characters shown as `\r`, `\n`)
   - `RX TCP` — a command line received on the TCP interface (port 4533)
   - `TX SER` / `TX TCP` — a reply written to that interface
   - `TX DROP` — a reply was generated but had no delivery target (e.g. no TCP client connected); it was **not** sent anywhere

- `TX CON` — console text printed to the serial wire (e.g. TCP client connect/disconnect notices)
- `[OVERFLOW]` / `[STALL: …]` — a received line was too long or never terminated; it was discarded

3. Reproduce the odd behaviour with your tracking software running.
4. Click **Download log** in the debug tab to save the capture as a `.log` file. Share the file when reporting a problem.
5. Close the debug tab to stop logging.

Notes:

- Logging is stored in RAM only — **nothing is written to the device's flash**, so there is no flash-wear concern. Capture starts when the debug tab polls the device and stops automatically about 5 seconds after the tab is closed.
- Each line carries `[<milliseconds since boot>]` plus a wall-clock timestamp (`YYYY-MM-DD HH:MM:SS.mmm`) once the device has a time reference. The device has no battery-backed clock, so the reference is obtained either way:
  - **Automatic (NTP)** — while connected to a router with internet access, the device syncs from NTP within about a minute of connecting; no action needed.
  - **Manual** — click **Sync time** in the debug tab. This sends your computer's clock and timezone to the device and works even in AP mode (where NTP cannot reach the device). Your UTC offset is remembered across reboots. Until a reference is set, lines show only the boot-milliseconds prefix, which still lets you measure gaps between a command and its reply. The downloaded log's header states the timestamp format, the UTC offset in use, and the time source.
- The log holds the most recent 96 lines; for longer sessions, download the log periodically and stitch the files together.

8.5 Configuring via OLED Menus

- **Protocol**: From main screen, cycle to "SETUP" → press Btn2 → select "PC Interface" → select "Protocol" → choose GS-232A/GS-232B/Easycom.
- **Hamlib Server**: SETUP → "Network" → select "Hamlib Server" → toggle Yes/No.
- **Rotator Variant**: SETUP → select "Rot Variant" → choose N360/S360/N450/S450.
- **ELF Mode**: From main screen, cycle to "ELF" → press Btn2 → toggle On/Off → respond to "Elevate To?" prompt.
- **WiFi Credentials (Captive Portal)**: SETUP → "Network" → "WiFi Mode" → "Station" → "Set WiFi Creds" → connect phone/laptop to AZELCTRL_XXXX → open browser to configure → press Btn1 to return.
- **Review Saved Credentials**: SETUP → "Network" → "WiFi Mode" → "Station" → "Review Creds" to view saved SSID and masked password.
- **Connect/Disconnect WiFi**: SETUP → "Network" → "WiFi Mode" → "Station" → scroll to "Connect WiFi" → press Btn2. Label changes to "Disconnect WiFi" when connected.
- **Show IP Address**: SETUP → "Network" → "WiFi Mode" → "Station" → "Show IP".
- **AP Mode (Field)**: SETUP → "Network" → "WiFi Mode" → "Station" → "AP Mode (Field)" — starts AP mode immediately without changing the saved WiFi mode. The "AP Mode" menu also provides "AP Status" and "Regen AP Password" options.

- **Regen AP Password**: SETUP → "Network" → "WiFi Mode" → "Access Point" → "Regen AP Password" → press Btn1 repeatedly for entropy → note the new password.
- **Calibrate Pots**: SETUP → "Rot Calibrate" → select Azimuth or Elevation → follow the sweep prompts. See section 6.6.
- **Skip Calibration**: SETUP → "Rot Calibrate" → "Skip (Az=El=3.2V)" — sets fixed 0–3.2V range without sweeping.
- **Wipe All Settings**: SETUP → "Wipe all settings" → "Continue" to erase all EEPROM data and reboot.

9. Troubleshooting

9.1 OLED Shows Nothing

Possible Cause	Solution
I2C connection loose	Check SDA=D6 and SCL=D7 connections
OLED not powered	Verify 3.3V and GND connections
Wrong I2C address	OLED should be at address 0x3C
OLED mounted incorrectly	Ensure screen faces outward and pins are properly seated

9.2 Motors Not Moving

Possible Cause	Solution
Motor enable pins not connected	Verify D1 (AZ enable) and D2 (EL enable) on shield
Motor shield not powered	Check VM connection to Yaesu +12V
Direction-reverse pause active	Wait for safety pause countdown to complete
Yaesu controller not powered	Verify Yaesu controller is on and receiving power
DIN connector loose	Check all terminal screws are tight

9.3 ADS1115 Not Found

Possible Cause	Solution
I2C bus conflict	Ensure no devices share conflicting addresses
Wiring issue	Check A0 and A1 connections to Yaesu feedback pins
Wrong I2C address	ADS1115 default address is 0x48; check address jumpers
Voltage too high	Verify Yaesu trim pots are adjusted to ≤3.3V

9.4 WiFi Connectivity Issues

Possible Cause	Solution
Wrong SSID/password	Reconfigure via captive portal (select "Set via HTTP")
Signal too weak	Move AZELCtrl closer to access point
IP address conflict	Check router DHCP lease table
ESP8266 not connecting	Check OLED for error status code

9.5 Position Readings Incorrect

Possible Cause	Solution
Potentiometers not calibrated	Run calibration procedure in OLED menu
Wrong rotator model selected	Verify Rotator Variant matches your hardware
Trim pots drifted	Re-adjust Yaesu controller trim pots
Loose wiring	Check ADS1115 connections to Yaesu feedback pins

9.6 Tracking Software Connection Refused

Possible Cause	Solution
AZELCtrl not connected to WiFi	Check OLED for WiFi status
Wrong port number	AZELCtrl listens on port 4533
Wrong protocol	Match Hamlib Server setting to software expectations
Firewall blocking	Check network firewall rules

9.7 Web Interface Issues

Possible Cause	Solution
Blank page after firmware flash	No HTML uploaded yet — upload `index.html.gz` via Settings tab
Blank page or `ERR_CONTENT_DECODING_FAILED`	`index.html.gz` was double-compressed — upload the `.gz` file, not the raw `.html`
Chrome shows `ERR_CONNECTION_RESET` repeatedly	ESP8266 TCP backlog fills up — use Firefox, or go to `chrome://flags/#unsafely-treat-insecure-origin-as-secure` and add the ESP's IP
Browser shows old version	Hard refresh (Cmd+Shift+R / Ctrl+F5)
Web page loads but map has no coastlines	`world_coastline.js` not uploaded — upload via Settings tab
Favicon not showing in browser tab	Use Chrome/Firefox, or clear browser cache
Tracking stops or becomes unreliable after switching to another browser tab	Browser froze the background tab — ensure **Background Keep-Alive** is enabled in Settings → Tracking Preferences (on by default); tracking resynchronises automatically when you return to the tab
External tracking software misbehaves over USB serial or TCP	Use **Debug USB Log** (Settings → PC Interface) to capture the exact protocol exchange and share the downloaded log with the developer
Web UI shows Az/El of 0 while the OLED shows a fixed position	The position ADC (I2C) is not being detected — open `http://<unit-IP>/diag` in a browser; if it reports `"ads":0`, reseat the OLED/ADC board connector (SDA/SCL lines) and power-cycle the unit. A failed I2C connection also freezes the OLED on its last frame, which is why the display can look "alive" while reporting nothing
Web UI shows a ROTOR OFFLINE banner / OLED shows ROTOR OFFLINE	The Yaesu controller is unplugged or unpowered (its +12 V rail is gone) — reconnect the controller and verify power. If the detection divider (1.3 MΩ + 200 kΩ, §3.3) is not fitted yet, the reading is meaningless: disable **Rotor Power Detection** in Settings until the resistors are installed

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10. Appendices

10.1 Yaesu DIN-8 Pinout Reference

Pin	Name	Function	Signal Type
1	EL Feedback	Elevation position voltage	Analog output (0–3.3V after calibration)
2	AZ CW	Azimuth clockwise control	Digital input (ground to activate)
3	EL Down	Elevation down control	Digital input (ground to activate)
4	AZ CCW	Azimuth counter-clockwise control	Digital input (ground to activate)
5	EL Up	Elevation up control	Digital input (ground to activate)
6	AZ Feedback	Azimuth position voltage	Analog output (0–3.3V after calibration)
7	+13.5V Out	Power output from controller	DC 12–14V (used for VM)
8	GND	Common ground	Ground reference

10.2 ESP8266 Pin Mapping Table

NodeMCU Label	GPIO	Function in AZELCtrl
D0	GPIO16	Unused (tricky for buttons)
D1	GPIO5	Motor A Enable (Azimuth)
D2	GPIO4	Motor B Enable (Elevation)
D3	GPIO0	Motor A Direction (Azimuth CW/CCW)
D4	GPIO2	Motor B Direction (Elevation Up/Down)
D5	GPIO14	Button 2 (active LOW)
D6	GPIO12	I2C SDA
D7	GPIO13	I2C SCL
D8	GPIO15	Button 1 (active HIGH)
TX	GPIO1	USB Serial TX
RX	GPIO3	USB Serial RX
A0	ADC0	Analog input (not used)

10.3 Protocol Command Reference

GS-232B Commands

Command	Description	Example	Response
`C`	Query position (azimuth, elevation)	`C\r`	`C125.45\r` (Az=125°, El=45°)
`C1`	Position with rate of change	`C1\r`	`AZ=125 EL=045 RATE=00.0\r`
`C2`	Position with status	`C2\r`	`AZ=125 EL=045\r`
`C3`	Position with full status	`C3\r`	`AZ=125 EL=045 STATUS=0\r`
`H`	Query azimuth heading	`H\r`	`H125\r`

Command	Description	Example	Response
`Hnnn`	Set azimuth to nnn°	`H180\r`	`OK\r`
`Mnnn`	Move azimuth to nnn°	`M180\r`	`OK\r`
`Wnnn eee`	Move to azimuth nnn, elevation eee	`W180 045\r`	`OK\r`
`V`	Query elevation	`V\r`	`V45\r`
`Vnnn`	Set elevation to nnn°	`V90\r`	`OK\r`
`L`	Azimuth left (CCW)	`L\r`	`OK\r`
`R`	Azimuth right (CW)	`R\r`	`OK\r`
`U`	Elevation up	`U\r`	`OK\r`
`D`	Elevation down	`D\r`	`OK\r`
`S`	Stop all movement	`S\r`	`OK\r`
`A`	All stop (alternate)	`A\r`	`OK\r`
`E`	Stop elevation only	`E\r`	`OK\r`
`Z`	Stop azimuth only	`Z\r`	`OK\r`
`Xnn`	Set speed (1=slow, 4=fast)	`X4\r`	`OK\r`

Easycom (EasyComm 2/3) Commands

Command	Description	Example	Response
`AZ`	Query azimuth	`AZ\r`	`AZ125.0\r`
`AZ xxx.x`	Set azimuth	`AZ 180.0\r`	`OK\r`
`EL`	Query elevation	`EL\r`	`EL45.0\r`
`EL yy.y`	Set elevation	`EL 90.0\r`	`OK\r`
`ML`	Move azimuth left (CCW)	`ML\r`	`OK\r`
`MR`	Move azimuth right (CW)	`MR\r`	`OK\r`
`MU`	Move elevation up	`MU\r`	`OK\r`
`MD`	Move elevation down	`MD\r`	`OK\r`
`S`	All stop	`S\r`	`OK\r`
`SA`	Stop azimuth only	`SA\r`	`OK\r`
`SE`	Stop elevation only	`SE\r`	`OK\r`
`VL`	Velocity — move left	`VL\r`	`OK\r`
`VR`	Velocity — move right	`VR\r`	`OK\r`
`VU`	Velocity — move up	`VU\r`	`OK\r`
`VD`	Velocity — move down	`VD\r`	`OK\r`
`VE`	Query firmware version	`VE\r`	`VEYCINT/1.0\r`
`GS`	Get status (1=Idle, 2=Moving)	`GS\r`	`GS1\r`
`AO`	AOS acknowledge	`AO\r`	`OK\r`
`LO`	LOS acknowledge	`LO\r`	`OK\r`

Command	Description	Example	Response
`\?`	Backslash query (rotctl identification)	`\?\r`	`\?EASYCOM,ROTATOR\r`

rotctl Commands

Command	Description	Example	Response
`p`	Query position	`p\r\n`	`125\n45\nRPRT 0\n`
`P az el`	Move to position	`P 180 45\r\n`	`RPRT 0\n`
`S`	Stop	`S\r\n`	`RPRT 0\n`

10.4 EEPROM Layout

Bytes	Size	Content	Magic Byte
0–66	67	WiFi SSID (data at 1–32) and password (data at 34–65)	0xAA at byte 66
67–83	17	Calibration min/max voltages (4× float)	0xCA at byte 83
84–85	2	WiFi mode (STA/AP/Disabled)	0x4D at byte 85
86–87	2	Network settings (Hamlib Server toggle)	0x4E at byte 87
88–89	2	Rotator model (N360/S360/N450/S450)	0x5A at byte 89
90	1	ELF mode enabled/disabled	—
91–99	9	AP password (8 chars)	0x41 ('A') at byte 99
100–104	5	Protocol settings — baud index, show commands, reverse delay, protocol variant	0xB7 at byte 102

End of Document

AZELCtrl — Built for amateur satellite operators, by amateur satellite operators.

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